DATA ANALYST - SECURITISATION AI AGENT

Category: Financial Data Analytics | Risk Assessment | Securitisation | DAX

Timeline: 15 Days | AI-Accelerated Innovation & Research Project

Target: Undergraduate / MBA / CA (Inter)

TASK:

Build securitisation risk assessment dashboards using the most complex DAX expressions for Microsoft Power BI, Analysis Services, and Power Pivot. Work with assumed bank data from SAP/ERP systems covering auto loans, mortgages, and structured credit. Develop IFRS 9 ECL models, waterfall calculations, stress testing frameworks, and investor reporting solutions. Design a gamified Securitisation Risk Arena simulation platform e.g. analyse real-world securitisation crises through a data analytics lens.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

PART E: NO-CODE TOOLS & PLATFORMS GUIDE

PART F: SUBMISSION PROTOCOL

Important Note:

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EXECUTIVE SUMMARY

Securitisation AI Agent is an enterprise-scale financial analytics project where data analyst interns build sophisticated risk assessment dashboards for securitised asset portfolios. Working with assumed data extracted from SAP/ERP systems of a financial institution, participants develop the most complex DAX (Data Analysis Expressions) formulations used across Microsoft data platforms including Power BI, Analysis Services, and Power Pivot in Excel.

The project transforms raw securitisation data - including mortgage-backed securities (MBS), asset-backed securities (ABS), collateralised loan obligations (CLOs), and collateralised debt obligations (CDOs) - into actionable risk intelligence through advanced DAX measures that would typically require years of experience to master. The project spans six comprehensive parts: Part A provides deep-dive training material on securitisation fundamentals and DAX analytics across all major asset classes; Part B presents the Securitisation Risk Arena gamified simulation business concept; Part C applies the analytics framework to real-world crises including the 2008 Global Financial Crisis and the IL&FS collapse; Part D provides a detailed 15-day day-by-day development methodology; Part E guides interns to the best no-code tools and regulatory data sources; and Part F specifies the 1000-point assessment criteria with detailed rubrics.

The core challenge students must solve: How does a data analyst transform raw loan-level data from SAP/ERP systems into the comprehensive suite of risk metrics that credit committees, rating agencies, investors, and regulators demand? The answer lies in mastering advanced DAX patterns - CALCULATE with complex filter modifications, iterator functions for weighted computations, virtual tables for dynamic stratification, time intelligence for trend analysis, and context transition for portfolio-level aggregation - combined with deep domain knowledge of securitisation structures, waterfall mechanics, IFRS 9 ECL computation, and regulatory reporting requirements.

Students will also design Securitisation Risk Arena - a browser-based gamified simulation platform that trains data analysts through realistic, time-pressured, scenario-driven challenges covering the full spectrum of securitisation analytics: from building basic pool metrics to navigating crisis-mode stress testing under time pressure.

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PART A: TRAINING MATERIAL & LEARNING RESOURCES

SECTION A1: FOUNDATIONAL CONCEPTS

A1.1 Securitisation Fundamentals

What is Securitisation?

Securitisation is the process of pooling various types of financial assets (loans, receivables, mortgages) and converting them into tradeable securities backed by the cash flows from those assets. The process involves transferring assets from an originator to a Special Purpose Vehicle (SPV), which issues securities (tranches) with varying risk-return profiles to investors.

Key Structures:

- Pass-Through Securities: Cash flows from underlying assets pass directly to investors proportionally
- Pay-Through Securities (CMOs): Cash flows are restructured into sequential tranches with different maturities
- Credit-Tranched Securities (CDOs/CLOs): Cash flows allocated by seniority-senior tranches absorb losses last, equity tranches absorb first

A1.2 DAX Fundamentals for Securitisation

Evaluation Context - The Foundation of DAX:

Every DAX expression evaluates within a context. Understanding context is the single most critical skill.

Filter Context:

```
Total Outstanding Balance =
SUM ( Loans\[CurrentBalance\] )
```

Row Context (Iterator):

```
Weighted Average LTV =
DIVIDE (
SUMX ( Loans, Loans\[CurrentBalance\] \* Loans\[LTV_Ratio\] ),
SUM ( Loans\[CurrentBalance\] )
)
```

Context Transition:

```
Portfolio ECL =
SUMX (
Loans,
CALCULATE ( \[PD\] \* \[LGD\] \* Loans\[EAD\] )
)
```

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SECTION A2: SECURITISATION DATA REQUIREMENTS

This section provides comprehensive coverage of data requirements across all major securitisation asset classes, as mandated by rating agencies (S&P Global Ratings, Moody's Investors Service, Fitch Ratings, and Indian agencies CRISIL and ICRA) and required by institutional investors. Understanding these data fields is critical for building accurate DAX models and Power BI dashboards.

A2.1 Auto Loan ABS - Data Requirements

Auto loan asset-backed securities are among the most common securitisation structures globally. Rating agencies require granular loan-level data to assess credit risk, prepayment risk, and residual value risk. The following data elements are mandatory for new issue ratings and ongoing surveillance.

A. Loan-Level Data Tape (Rating Agency Requirement)

The loan-level data tape is the single most important document in any auto loan securitisation. S&P, Moody's, and Fitch all require this tape at cutoff for initial rating and monthly for surveillance. Each row represents one loan in the pool.

Loan Identification Fields:

Field Name	Data Type	Description	Rating Agency Use
Loan ID	VARCHAR(20)	Unique loan identifier	De-duplication, tracking
Pool ID	VARCHAR(15)	Securitisation deal identifier	Pool allocation
Originator ID	VARCHAR(10)	Lending institution code	Originator concentration
Servicer ID	VARCHAR(10)	Current servicer code	Servicing risk assessment
Account Status	VARCHAR(20)	Current/Delinquent/Default	Portfolio health

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Origination Data Fields:

Field Name	Data Type	Description	Rating Agency Use
Origination Date	DATE	Date loan was funded	Seasoning, vintage analysis
Original Balance	DECIMAL(15,2)	Original loan amount	Pool factor calculation
Original Term	INT	Loan term in months	Weighted avg maturity
Interest Rate	DECIMAL(5,2)	Contract coupon rate	WAC, excess spread
Rate Type	VARCHAR(10)	Fixed/Floating/Hybrid	Interest rate risk
Origination Channel	VARCHAR(20)	Dealer/Direct/Online/DSA	Channel risk analysis
Loan Purpose	VARCHAR(30)	New/Used/Refinance	Collateral risk

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Vehicle/Collateral Data Fields:

Field Name	Data Type	Description	Rating Agency Use
Vehicle Make	VARCHAR(30)	Manufacturer name	Residual value curves
Vehicle Model	VARCHAR(30)	Model name	Depreciation analysis
Vehicle Year	INT	Manufacturing year	Age-based default curves
Vehicle Type	VARCHAR(20)	Sedan/SUV/Hatchback/ Truck	Segment risk factors
New/Used Flag	BOOLEAN	New or pre-owned	LGD differentiation
Original Vehicle Value	DECIMAL(15,2)	Value at origination	LTV at origination
Current Vehicle Value	DECIMAL(15,2)	Current estimated value	Current LTV
Mileage at Origination	INT	Odometer reading (km)	Depreciation input
Vehicle Registration State	VARCHAR(30)	State of registration	Geographic concentration

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Borrower Data Fields:

Field Name	Data Type	Description	Rating Agency Use
CIBIL Score at Origination	INT	Credit score at loan booking	PD model input
CIBIL Score Current	INT	Latest credit score	Credit migration
Employment Type	VARCHAR(20)	Salaried/Self-Employed/Business	Income stability
Annual Income	DECIMAL(15,2)	Verified annual income	DTI calculation
DTI Ratio	DECIMAL(5,4)	Debt-to-income ratio	Affordability assessment
Borrower Age	INT	Age at origination	Demographic risk
Geographic Region	VARCHAR(20)	State/Region	Concentration risk
Years at Current Employment	INT	Employment tenure	Stability indicator

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Performance Data Fields (Monthly Updated):

Field Name	Data Type	Description	Rating Agency Use
Reporting Date	DATE	Snapshot date	Time series tracking
Current Balance	DECIMAL(15,2)	Outstanding principal	Pool factor
Scheduled Payment	DECIMAL(12,2)	Monthly EMI due	Collection rate
Actual Payment	DECIMAL(12,2)	Amount collected	Delinquency calc
Delinquency Status	VARCHAR(20)	Current/30/60/90/Default	Delinquency pipeline
Days Past Due	INT	Number of days delinquent	Bucket classification
Cumulative Loss	DECIMAL(15,2)	Total realised losses	Loss curve
Recovery Amount	DECIMAL(15,2)	Recovered from defaults	Recovery rate
Prepayment Amount	DECIMAL(15,2)	Voluntary prepayment	CPR/ABS calculation
Modification Flag	BOOLEAN	Loan restructured flag	Performance exclusion
IFRS 9 Stage	INT	1/2/3 staging	ECL provision bucket

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B. Pool-Level Statistics Required by Rating Agencies

In addition to loan-level tapes, rating agencies require the following pool-level summary statistics. These are calculated using DAX measures in the Power BI model:

Pool Composition Metrics:

- Weighted Average Coupon (WAC): Interest rate weighted by current balance
- Weighted Average Maturity (WAM): Remaining term weighted by current balance
- Weighted Average Loan Age (WALA): Months on book weighted by current balance
- Weighted Average LTV: Loan-to-value ratio weighted by current balance
- Weighted Average CIBIL Score: Credit score weighted by current balance
- Weighted Average DTI: Debt-to-income weighted by current balance

DAX Implementation - All Weighted Averages:

```
WAC = DIVIDE(SUMX(Loans,
Loans\[InterestRate\]\*Loans\[CurrentBalance\]),
SUM(Loans\[CurrentBalance\]))
WAM = DIVIDE(SUMX(Loans,
Loans\[RemainingTerm\]\*Loans\[CurrentBalance\]),
SUM(Loans\[CurrentBalance\]))
WALA = DIVIDE(SUMX(Loans,
Loans\[MonthsOnBook\]\*Loans\[CurrentBalance\]),
SUM(Loans\[CurrentBalance\]))
Weighted Avg LTV =
DIVIDE(
SUMX(FactLoanPerformance,
FactLoanPerformance\[LTV_Current\]\*FactLoanPerformance\[Curre
ntBalance\]),
SUM(FactLoanPerformance\[CurrentBalance\])
)
```

Concentration Metrics:

- Top 10 Borrower Concentration: Percentage of pool held by 10 largest borrowers
- Geographic Concentration: Top 5 states by outstanding balance
- Originator Concentration: Breakdown by originating entity
- Vehicle Make Concentration: Exposure by manufacturer
- Herfindahl-Hirschman Index (HHI): Measure of portfolio diversification

```
HHI Index =
VAR TotalBal = CALCULATE(SUM(Loans\[CurrentBalance\]),
ALL(DimBorrower))
VAR BorrowerShares =
ADDCOLUMNS(
SUMMARIZE(Loans, DimBorrower\[BorrowerID\]),
\"Share\",      DIVIDE(CALCULATE(SUM(Loans\[CurrentBalance\])),
TotalBal)
)
RETURN SUMX(BorrowerShares, \[Share\]\^2)
Top N Borrower Exposure =
VAR TotalBal = CALCULATE(SUM(Loans\[CurrentBalance\]),
ALL(DimBorrower))
VAR BorrowerRanked =
ADDCOLUMNS(
SUMMARIZE(Loans, DimBorrower\[BorrowerID\]),
\"Balance\", CALCULATE(SUM(Loans\[CurrentBalance\]))
)
VAR TopN = TOPN(10, BorrowerRanked, \[Balance\], DESC)
RETURN DIVIDE(SUMX(TopN, \[Balance\]), TotalBal)
```

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C. S&P Global - Specific Auto ABS Requirements

S&P Global Ratings uses its LEVELS (Loan Evaluation and Estimate of Loss System) model for auto ABS. The model requires specific data inputs beyond the standard tape:

LEVELS Model Inputs:

- Base Case Loss Expectation: Derived from historical loss curves of the originator
- Loss Timing Curve: Monthly distribution of expected losses
- Recovery Lag: Average time from default to recovery (typically 3-6 months for auto)
- Voluntary Prepayment Speed: Expressed as Absolute Prepayment Speed (ABS) or Conditional Prepayment Rate (CPR)
- Recovery Rate by LTV Band: Different recovery assumptions for high-LTV vs low-LTV
- Stress Multiples by Rating Level: AAA = 3.0x-4.0x base loss, AA = 2.5x-3.0x, A = 2.0x-2.5x

DAX for S&P Stress Scenarios:

```
SP Stressed Loss AAA =
VAR BaseLoss = \[Base Case Expected Loss\]
VAR StressMultiple = 3.5 // AAA stress factor
VAR StressedLoss = BaseLoss \* StressMultiple
VAR CreditEnhancement = \[Senior Tranche CE Percentage\]
RETURN
IF(StressedLoss  \>  CreditEnhancement,  \"INSUFFICIENT  CE\",
\"ADEQUATE
CE\")
```

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D. Moody's - Specific Auto ABS Requirements

Moody's uses its Idealised Default Table and MILAN (Moody's Individual Loan Analysis) framework. Key additional data requirements:

- Lognormal Loss Distribution Parameters: Mean and standard deviation of loss
- Correlation Assumptions: Intra-pool asset correlation (typically 3-8% for auto)
- Seasoning Adjustment: Performance credit for seasoned pools
- Originator Assessment: Qualitative score of originator's underwriting standards
- Servicer Quality Assessment: Score of servicer's collection capabilities

E. Fitch Ratings - Specific Auto ABS Requirements

Fitch uses a multi-step approach requiring:

- Expected Cumulative Net Loss (CNL): Base case loss estimate
- Loss Timing Distribution: Front-loaded vs back-loaded loss patterns
- Rating Factor Multiples: Stress multiples applied to base case (AAAsf = 5.0x-6.0x for subprime, 3.0x-4.0x for prime)
- Excess Spread Analysis: Monthly excess spread available to absorb losses
- Liquidity Assessment: Ability of structure to meet payment obligations during stress

F. Indian Rating Agencies - CRISIL and ICRA Requirements

For Indian auto loan securitisations, CRISIL and ICRA require additional India-specific data:

- RBI Priority Sector Classification: Whether loans qualify under PSL guidelines
- GST Registration of Borrower: For commercial vehicle loans
- Aadhaar Verification Status: KYC completeness
- NACH Mandate Status: Auto-debit consent for EMI collection
- Insurance Coverage: Comprehensive vs third-party vs uninsured
- Hypothecation Registration: RC book endorsement status
- Loan Restructuring Under RBI Moratorium: Covid moratorium status history

DAX for Indian Regulatory Metrics:

```
NACH Mandate Coverage =
DIVIDE(
CALCULATE(SUM(Loans\[CurrentBalance\]),
Loans\[NACH_Status\]=\"Active\"),
SUM(Loans\[CurrentBalance\])
)
Insurance Coverage Rate =
DIVIDE(
CALCULATE(COUNTROWS(Loans),
Loans\[InsuranceType\]\<\>\"Nil\"),
COUNTROWS(Loans)
)
```

A2.2 Residential Mortgage-Backed Securities (RMBS) - Data Requirements

RMBS securitisations are the most data-intensive asset class. Rating agencies require extensive property-level, borrower-level, and loan-level data to assess credit risk, prepayment risk, and market value decline risk.

A. RMBS Loan-Level Data Tape

Mortgage-Specific Fields (beyond generic loan fields):

Field Name	Data Type	Description	Rating Agency Use
Property Type	VARCHAR(20)	Flat/House/Plot/Commercial	Property risk segmentation
Property City	VARCHAR(30)	City/town of property	Metro/Tier-2/Tier-3 analysis
Property PIN Code	VARCHAR(6)	Postal code	Micro-market risk
Property Area (sq ft)	INT	Built-up area	Per sq ft valuation
Original Valuation	DECIMAL(15,2)	Appraised value at origination	LTV at origination
Current Valuation	DECIMAL(15,2)	Most recent appraisal	Current LTV, MVD risk

Valuation Date	DATE	Date of latest appraisal	Staleness of valuation
Valuation Method	VARCHAR(20)	Desktop/Drive-by/Full	Valuation reliability
CERSAI Registration	BOOLEAN	Registered with CERSAI	Legal enforceability
Title Insurance	BOOLEAN	Title insurance obtained	Legal risk mitigation
Occupancy Status	VARCHAR(20)	Owner-occupied/Rented/Vacant	Default probability input
Co-Borrower Flag	BOOLEAN	Loan has co-applicant	Income stability
PMAY Subsidy	BOOLEAN	Pradhan Mantri Awas Yojana	Government guarantee
DSCR	DECIMAL(5,2)	Debt Service Coverage Ratio	Affordability metric

RMBS-Specific Rating Agency Requirements:

- Market Value Decline (MVD) Assumptions: S&P uses Quick Sale Adjustment (QSA) of 15-20% for India
- Foreclosure Timeline: Average time from default to property liquidation (India: 18-36 months under SARFAESI)
- SARFAESI Applicability: Whether lender can invoke SARFAESI Act for recovery
- Property Price Index (PPI): Regional house price appreciation/depreciation trends
- Rental Yield Data: For investment properties, used in DSCR calculations

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DAX for RMBS-Specific Metrics:

```
Market Value Decline Stressed LTV =
VAR CurrentLTV = \[Weighted Avg LTV\]
VAR MVD_Haircut = 0.20 // 20% market value decline
VAR StressedPropertyValue =
SUMX(Loans, Loans\[CurrentPropertyValue\] \* (1 -
MVD_Haircut))
VAR TotalBalance = SUM(Loans\[CurrentBalance\])
RETURN DIVIDE(TotalBalance, StressedPropertyValue)
Foreclosure Period Loss =
VAR AvgForeclosurePeriod = 24 // months
VAR MonthlyCarryCost = 0.005 // 50bps monthly
VAR CarryCostLoss =
SUMX(
FILTER(Loans, Loans\[IsDefaulted\]),
Loans\[CurrentBalance\] \* MonthlyCarryCost \*
AvgForeclosurePeriod
)
RETURN CarryCostLoss
```

B. RMBS Pool Stratification Tables

Rating agencies require the following stratification tables. Each stratification shows loan count, current balance, percentage of pool, weighted average coupon, and weighted average LTV:

1. LTV Distribution: 0-50%, 50-60%, 60-70%, 70-75%, 75-80%, 80-85%, 85-90%, 90%+
2. CIBIL Score Distribution: \<600, 600-650, 650-700, 700-750, 750-800, 800+
3. Loan Size Distribution: \<5L, 5-10L, 10-20L, 20-50L, 50L-1Cr, 1Cr+
4. Geographic Distribution: By State and by Metro/Tier-2/Tier-3 classification
5. Seasoning Distribution: 0-6M, 6-12M, 12-24M, 24-36M, 36-60M, 60M+
6. Property Type Distribution: Flat, Independent House, Row House, Plot + Construction
7. Employment Type: Salaried, Self-Employed Professional, Self-Employed Business
8. Interest Rate Distribution: By 50bps bands
9. Original Term Distribution: By 12-month bands
10. DTI Distribution: \<20%, 20-30%, 30-40%, 40-50%, 50%+

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DAX for Dynamic Stratification:

```
// Generic stratification measure - works with any dimension
Strat % of Pool =
DIVIDE(
SUM(FactLoanPerformance\[CurrentBalance\]),
CALCULATE(SUM(FactLoanPerformance\[CurrentBalance\]),
REMOVEFILTERS())
)
Strat Loan Count =
DISTINCTCOUNT(FactLoanPerformance\[LoanID\])
Strat WAC =
DIVIDE(
SUMX(FactLoanPerformance,
FactLoanPerformance\[InterestRate\]\*FactLoanPerformance\[Curr
entBalance\]),
SUM(FactLoanPerformance\[CurrentBalance\])
)
```

A2.3 Collateralised Loan Obligations (CLO) - Data Requirements

CLOs securitise portfolios of leveraged loans (senior secured corporate loans). The data requirements are significantly different from consumer ABS due to the corporate nature of underlying assets.

A. Loan-Level Data for CLO Collateral

Field Name	Data Type	Description	Rating Agency Use
Obligor Name	VARCHAR(100)	Corporate borrower name	Concentration limits
Industry Code	VARCHAR(10)	GICS/ICB industry code	Industry concentration
Moody's CFR	VARCHAR(5)	Corporate Family Rating	Default probability
S&P Issuer Rating	VARCHAR(5)	Issuer credit rating	Rating migration risk
Facility Type	VARCHAR(20)	Term Loan A/B, Revolver	Seniority

Seniority	VARCHAR(20)	First Lien/Second Lien	Recovery rate
EBITDA	DECIMAL(15,2)	Trailing 12M EBITDA	Leverage calculation
Total Debt	DECIMAL(15,2)	Total funded debt	Leverage ratio
Leverage Ratio	DECIMAL(5,2)	Debt/EBITDA	Credit quality
Interest Coverage	DECIMAL(5,2)	EBITDA/Interest	Debt service ability
Spread (bps)	INT	Margin over benchmark	Yield analysis
LIBOR/SOFR Floor	DECIMAL(5,2)	Rate floor	Excess spread
Maturity Date	DATE	Final maturity	WAL calculation
Country	VARCHAR(30)	Country of domicile	Sovereign risk
Cov-Lite Flag	BOOLEAN	Covenant-lite loan	Structural risk

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B. CLO Compliance Tests (Calculated in DAX)

These tests must be monitored monthly and displayed on dashboards:

```
// Weighted Average Spread Test
WAS Test =
VAR PortfolioWAS =
DIVIDE(SUMX(CLOLoans,
CLOLoans\[Spread_bps\]\*CLOLoans\[ParAmount\]),
SUM(CLOLoans\[ParAmount\]))
VAR WAS_Trigger = 350 // basis points
RETURN IF(PortfolioWAS \>= WAS_Trigger, \"PASS\", \"FAIL\")
// Weighted Average Rating Factor (WARF) Test
WARF Test =
VAR PortfolioWARF =
DIVIDE(
SUMX(CLOLoans, RELATED(RatingFactors\[MoodysRF\]) \*
CLOLoans\[ParAmount\]),
SUM(CLOLoans\[ParAmount\])
)
VAR WARF_Limit = 2800
RETURN IF(PortfolioWARF \<= WARF_Limit, \"PASS\", \"FAIL\")
// Diversity Score (Moody's)
Diversity Score =
VAR IndustryConcentrations =
ADDCOLUMNS(
SUMMARIZE(CLOLoans, CLOLoans\[IndustryCode\]),
\"Share\", DIVIDE(CALCULATE(SUM(CLOLoans\[ParAmount\])),
CALCULATE(SUM(CLOLoans\[ParAmount\]),ALL(CLOLoans)))
)
RETURN SUMX(IndustryConcentrations, MIN(\[Share\], 0.10/0.10))
// CCC Bucket Test
CCC Exposure =
DIVIDE(
CALCULATE(SUM(CLOLoans\[ParAmount\]), CLOLoans\[MoodysCFR\] IN
{\"Caa1\",\"Caa2\",\"Caa3\",\"Ca\",\"C\"}),
SUM(CLOLoans\[ParAmount\])
)
// Typically limited to 7.5% of portfolio
```

A2.4 Credit Card ABS - Data Requirements

Credit card securitisations are revolving structures (unlike amortising auto/mortgage pools). The data requirements reflect the revolving nature and the trust structure.

A. Account-Level Data Tape

Field Name	Data Type	Description	Rating Agency Use
Account ID	VARCHAR(20)	Unique card account number	Tracking
Account Open Date	DATE	Date card was issued	Seasoning
Credit Limit	DECIMAL(12,2)	Sanctioned credit limit	Utilisation analysis
Current Balance	DECIMAL(12,2)	Outstanding balance	Pool size
Minimum Payment Due	DECIMAL(10,2)	Monthly minimum payment	Payment rate
Payment Amount	DECIMAL(10,2)	Actual payment received	Payment rate calc
Purchase Volume	DECIMAL(12,2)	Monthly purchases	Trust yield
Cash Advance Volume	DECIMAL(12,2)	Cash withdrawals	Higher risk segment
APR – Purchase	DECIMAL(5,2)	Annual rate for purchases	Yield calculation
APR – Cash Advance	DECIMAL(5,2)	Rate for cash advances	Yield calculation
Delinquency Status	VARCHAR(20)	Current/30/60/90/Chargeoff	Loss measurement
Utilisation Rate	DECIMAL(5,4)	Balance / Credit Limit	Risk indicator

Revolving Flag	BOOLEAN	Carries balance month-to-month	Yield analysis

B. Credit Card Trust Performance Metrics

Monthly trust-level metrics required by investors:

```
// Monthly Payment Rate (MPR) - Key credit card metric
Monthly Payment Rate =
DIVIDE(
SUM(CardAccounts\[PaymentAmount\]),
CALCULATE(SUM(CardAccounts\[CurrentBalance\]),
PREVIOUSMONTH(DimDate\[Date\]))
)
// Excess Spread
Excess Spread =
\[Trust Yield\] - \[Investor Coupon\] - \[Servicing Fee\] -
\[Charge-off Rate\]
// Trust Yield
Trust Yield =
DIVIDE(
SUM(CardAccounts\[InterestCharged\]) +
SUM(CardAccounts\[FeeIncome\]),
CALCULATE(SUM(CardAccounts\[CurrentBalance\]),
PREVIOUSMONTH(DimDate\[Date\]))
) \* 12 // Annualised
// Three-Month Average Excess Spread (Early Amortisation
Trigger)
3M Avg Excess Spread =
```

```
AVERAGEX(
DATESINPERIOD(DimDate\[Date\], MAX(DimDate\[Date\]), -3,
MONTH),
CALCULATE(\[Excess Spread\])
)
```

Early amortisation is triggered if the three-month average excess spread falls below zero. This is a critical metric for credit card ABS investors.

A2.5 Commercial Mortgage-Backed Securities (CMBS) - Data Requirements

CMBS securitises commercial real estate loans. The data is property-focused rather than borrower-focused, with emphasis on cash flow analysis and property valuation.

A. Property-Level Data

Field Name	Data Type	Description	Rating Agency Use
Property Name	VARCHAR(100)	Name of commercial property	Identification
Property Type	VARCHAR(30)	Office/Retail/Industrial/Hotel	Sector risk
Net Operating Income	DECIMAL(15,2)	Annual NOI	DSCR calculation
Appraised Value	DECIMAL(15,2)	Most recent appraisal	LTV calculation
Occupancy Rate	DECIMAL(5,2)	Current occupied percentage	Cash flow stability
Gross Leasable Area	INT	Total leasable sq ft	Size metric
Major Tenant	VARCHAR(100)	Largest tenant by rent	Tenant concentration
WALT (Lease)	DECIMAL(5,2)	Weighted avg lease term	Cash flow duration

Cap Rate	DECIMAL(5,4)	Capitalisation rate	Valuation metric
DSCR	DECIMAL(5,2)	Debt Service Coverage Ratio	Primary credit metric
Year Built	INT	Construction year	Physical risk
Environmental Grade	VARCHAR(5)	LEED/GRIHA rating	ESG metric

CMBS-Specific DAX Measures:

```
// DSCR at Pool Level
Pool DSCR =
DIVIDE(
SUM(CMBSLoans\[NetOperatingIncome\]),
SUM(CMBSLoans\[AnnualDebtService\])
)
// LTV at Property Level
CMBS Weighted LTV =
DIVIDE(
SUM(CMBSLoans\[CurrentBalance\]),
SUMX(CMBSLoans, CMBSLoans\[AppraisedValue\])
)
// Maturity Default Risk (Balloon Risk)
Balloon Exposure =
CALCULATE(
SUM(CMBSLoans\[CurrentBalance\]),
CMBSLoans\[AmortisationType\] = \"Interest Only\"
\|\| CMBSLoans\[BalloonPaymentDue\] = TRUE
)
```

A2.6 Student Loan ABS - Data Requirements

Student loan securitisations in India (growing rapidly under the Vidya Lakshmi scheme) require unique data elements reflecting the education sector.

A. Student Loan Data Tape

Field Name	Data Type	Description	Rating Agency Use
Student ID	VARCHAR(20)	Unique student identifier	Tracking
Institution Name	VARCHAR(100)	College/university	Institution quality
Course Type	VARCHAR(30)	UG/PG/Professional/PhD	Employment probability
Course Duration	INT	Duration in months	Repayment start timing
Moratorium Period	INT	Months before repayment starts	Cash flow timing
Co-Borrower Income	DECIMAL(15,2)	Parent/guardian income	Affordability
Collateral Type	VARCHAR(30)	Property/FD/LIC/Unsecured	Recovery potential
Government Subsidy	BOOLEAN	Interest subsidy received	Cash flow impact
Employment Status	VARCHAR(20)	Employed/Unemployed/Studying	Repayment capacity
Starting Salary	DECIMAL(12,2)	First job salary (if employed)	Repayment capacity

Student loan ABS presents unique challenges for DAX modelling because of the moratorium period (no payments during study + 6-12 months after graduation). The delinquency pipeline must account for loans not yet in repayment.

A2.7 Investor Reporting Requirements - All Asset Classes

Institutional investors (pension funds, insurance companies, mutual funds, hedge funds) require standardised monthly reporting from the servicer/trustee. The Power BI dashboard must generate these reports with full DAX automation.

A. Monthly Servicer Report - Standard Sections

Section 1: Pool Summary

- Beginning-of-Period Pool Balance
- Scheduled Principal Collected
- Unscheduled Principal (Prepayments)
- Losses Realised
- Recoveries on Prior Losses
- End-of-Period Pool Balance
- Pool Factor (Current Balance / Original Balance)
- Number of Loans: Beginning, Added, Removed, Ending

```
Pool Factor =

DIVIDE(SUM(Loans\[CurrentBalance\]),

CALCULATE(SUM(Loans\[OriginalLoanAmount\]), ALL(DimDate)))
```

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Section 2: Delinquency and Default Report

Delinquency Bucket	# Loans	Balance (INR)	% of Pool	Migration
Current	###	###	##.#%	Cured from 30 DPD
1-29 DPD	###	###	##.#%	New delinquencies
30-59 DPD	###	###	##.#%	Rolled from 1-29
60-89 DPD	###	###	##.#%	Rolled from 30-59
90-119 DPD	###	###	##.#%	Rolled from 60-89
120+ DPD	###	###	##.#%	Rolled from 90-119
Default	###	###	##.#%	New defaults
Repossessed	###	###	##.#%	In recovery

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Section 3: Loss and Recovery Report

- Cumulative Gross Losses
- Cumulative Recoveries
- Cumulative Net Losses
- Annualised Net Loss Rate
- Recovery Rate (Recoveries / Gross Losses)
- Average Time to Recovery (months)

```
// Annualised Net Loss Rate
Annualised Net Loss Rate =
VAR MonthlyLoss = \[Net Loss This Month\]
VAR AvgBalance =
DIVIDE(
CALCULATE(SUM(Loans\[CurrentBalance\]),
PREVIOUSMONTH(DimDate\[Date\]))
\+ SUM(Loans\[CurrentBalance\]),
2
)
RETURN DIVIDE(MonthlyLoss, AvgBalance) \* 12
```

Section 4: Prepayment Report

- Monthly Prepayment Amount
- Single Month Mortality (SMM)
- Conditional Prepayment Rate (CPR) - Annualised
- Constant Prepayment Rate by Vintage

```
// Single Month Mortality
SMM =
DIVIDE(
SUM(Loans\[PrepaymentAmount\]),
CALCULATE(SUM(Loans\[CurrentBalance\]),
PREVIOUSMONTH(DimDate\[Date\]))
\- SUM(Loans\[ScheduledPrincipal\])
)
// Conditional Prepayment Rate (Annualised)
CPR = 1 - POWER(1 - \[SMM\], 12)
```

Section 5: Cash Flow Waterfall Report

- Total Collections (Interest + Principal)
- Less: Senior Fees (Trustee, Servicer, Rating Agency)
- Less: Senior Tranche Interest
- Less: Senior Tranche Principal
- Less: Mezzanine Tranche Interest (if OC/IC tests pass)
- Less: Mezzanine Tranche Principal
- Residual to Equity/Originator
- Reserve Account Balance

Section 6: Trigger/Covenant Status

Test	Trigger Level	Current Level	Status	Action
OC Test (Senior)	>=120%	###%	PASS/FAIL	Sequential pay if fail
OC Test (Mezz)	>=110%	###%	PASS/FAIL	Divert to deleverage
IC Test (Senior)	>=130%	###%	PASS/FAIL	Trap excess spread
IC Test (Mezz)	>=115%	###%	PASS/FAIL	Redirect cash flows
Delinquency Test	<=5.0%	###%	PASS/FAIL	Early amortisation
Cumulative Loss Test	<=3.0%	###%	PASS/FAIL	Accelerate paydown

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B. Investor Data Portals and EDGAR/SEBI Requirements

For Indian securitisations, SEBI requires issuers to provide investor data through designated portals. The following reporting frequencies apply:

Report	Frequency	Deadline	Recipient
Loan-Level Data Tape	Monthly	T+15 business days	Rating Agencies, Investors
Servicer Report	Monthly	T+15 business days	Trustee, Investors
Pool Performance Report	Monthly	T+15 business days	SEBI, Stock Exchange
Compliance Certificate	Quarterly	T+30 calendar days	Trustee, Rating Agencies
Annual Review	Annually	Within 6 months of FY end	Rating Agencies
Material Event Notice	As required	Within 2 business days	All parties

C. ESG Data Requirements (Emerging Standard)

Rating agencies increasingly require ESG-related data for securitisations. Power BI dashboards should include these emerging metrics:

Environmental Factors:

- Energy efficiency ratings of financed properties (for RMBS)
- Electric vehicle percentage (for auto ABS)
- Climate risk exposure (flood zone, wildfire zone, cyclone prone)

Social Factors:

- Affordable housing percentage (for RMBS)
- Financial inclusion metrics (rural vs urban borrowers)
- Borrower hardship program participation rate

Governance Factors:

- Originator compliance score
- Servicer complaint resolution rate
- Data quality and reporting timeliness

```
EV Percentage in Auto Pool =

DIVIDE(

CALCULATE(SUM(Loans\[CurrentBalance\]),

DimVehicle\[FuelType\]=\"Electric\"),

SUM(Loans\[CurrentBalance\])

)
```

A2.8 Days Past Due (DPD) Tables - Comprehensive Analysis

Days Past Due analysis is the single most scrutinised section of any securitisation performance report. Rating agencies, investors, trustees, and regulators all monitor DPD tables monthly to assess portfolio health, predict future losses, and determine whether structural triggers have been breached. This section provides exhaustive coverage of DPD data requirements and DAX implementations.

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A. DPD Bucket Definitions and Standards

The industry-standard DPD buckets used across all securitisation asset classes are defined below. Note that different jurisdictions and rating agencies may use slightly different definitions-the DAX model must be flexible enough to accommodate all:

DPD Bucket	Days Range	RBI Classification	IFRS 9 Stage	S&P Treatment	Moody's Treatment
Current	0 DPD	Standard	Stage 1	Performing	Performing
1-29 DPD	1-29 days	SMA-0	Stage 1	Performing	Performing
30-59 DPD	30-59 days	SMA-1	Stage 2	Delinquent	Delinquent
60-89 DPD	60-89 days	SMA-2	Stage 2	Delinquent	Delinquent
90-119 DPD	90-119 days	Sub-standard	Stage 3	Defaulted	Defaulted
120-149 DPD	120-149 days	Sub-standard	Stage 3	Defaulted	Defaulted
150-179 DPD	150-179 days	Doubtful	Stage 3	Defaulted	Defaulted
180-359 DPD	180-359 days	Doubtful	Stage 3	Defaulted	Charged-off
360+ DPD	360+ days	Loss	Stage 3	Loss	Loss

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RBI Special Mention Account (SMA) Classification:

The Reserve Bank of India requires banks to report SMA classifications as an early warning system. For securitised pools, these classifications feed directly into rating agency surveillance:

- SMA-0: Principal or interest payment overdue between 1-30 days
- SMA-1: Principal or interest payment overdue between 31-60 days
- SMA-2: Principal or interest payment overdue between 61-90 days
- NPA (Non-Performing Asset): Principal or interest overdue for more than 90 days

Note: Under RBI's revised framework (November 2021), the date of NPA classification is determined by the exact day count, not by the end of the month. This affects how DAX measures count DPD.

B. DPD Snapshot Table - Data Structure

The DPD snapshot table is a monthly time-series of delinquency states for every loan. This table is the foundation of all delinquency analysis and must be stored in the fact table:

Column Name	Data Type	Description	Example
SnapshotDate	DATE	Reporting period end date	2024-10-31
LoanID	VARCHAR(20)	Unique loan identifier	ZAAUTO2024-1-00001
DPD_Days	INT	Exact number of days past due	47
DPD_Bucket	VARCHAR(15)	Bucket classification	30-59 DPD
DPD_Bucket_Prior	VARCHAR(15)	Bucket in prior snapshot	1-29 DPD
CurrentBalance	DECIMAL(15,2)	Balance at snapshot	487,250.00
AmountOverdue	DECIMAL(12,2)	Total overdue amount	23,150.00
EMIsOverdue	INT	Number of EMIs missed	2
LastPaymentDate	DATE	Most recent payment received	2024-09-15
LastPaymentAmount	DECIMAL(12,2)	Amount of last payment	11,575.00

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CureFlag	BOOLEAN	Cured from delinquency this period	FALSE
RollFlag	VARCHAR(10)	Roll direction: Worse/Better/Same	Worse
RepossessionFlag	BOOLEAN	Vehicle/property repossessed	FALSE
WriteOffFlag	BOOLEAN	Loan written off this period	FALSE
RBI_SMA_Class	VARCHAR(5)	SMA-0/SMA-1/SMA-2/NPA	SMA-1

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C. DPD Distribution Report - Core DAX Measures

The following DAX measures generate the complete DPD distribution report required by investors. Each measure produces both loan-count and balance-weighted views:

Measure Group 1: DPD Bucket Counts and Balances

```
// Loans in each DPD bucket
DPD Loan Count =
COUNTROWS ( FactLoanPerformance )
DPD Balance =
SUM ( FactLoanPerformance\[CurrentBalance\] )
DPD Percentage of Pool (by Balance) =
DIVIDE (
\[DPD Balance\],
CALCULATE ( \[DPD Balance\], REMOVEFILTERS (
FactLoanPerformance\[DPD_Bucket\] ) )
)
DPD Percentage of Pool (by Count) =
DIVIDE (
\[DPD Loan Count\],
CALCULATE ( \[DPD Loan Count\], REMOVEFILTERS (
FactLoanPerformance\[DPD_Bucket\] ) )
)
```

Measure Group 2: Specific Bucket Measures (for KPI Cards)

```
// Current (0 DPD) - Performing
Current Balance =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] = 0
)
// 30+ DPD - Total Delinquent (Rating Agency Key Metric)
30Plus DPD Balance =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 30
)
30Plus DPD Rate =
DIVIDE ( \[30Plus DPD Balance\], SUM (
FactLoanPerformance\[CurrentBalance\] ) )
// 60+ DPD - Seriously Delinquent
60Plus DPD Balance =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 60
)
60Plus DPD Rate =
DIVIDE ( \[60Plus DPD Balance\], SUM (
FactLoanPerformance\[CurrentBalance\] ) )
// 90+ DPD - Default/NPA (RBI Definition)
```

```
90Plus DPD Balance =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 90
)
NPA Rate (RBI) =
DIVIDE ( \[90Plus DPD Balance\], SUM (
FactLoanPerformance\[CurrentBalance\] ) )
```

Measure Group 3: DPD Trend Over Time

```
// 30+ DPD Rate as of any selected month (for time-series chart)
30Plus DPD Rate by Month =
VAR SelectedMonth = MAX ( DimDate\[Date\] )
VAR TotalBal =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
DimDate\[Date\] = SelectedMonth
)
VAR DelinqBal =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 30,
DimDate\[Date\] = SelectedMonth
)
RETURN DIVIDE ( DelinqBal, TotalBal )
// Rolling 3-Month Average DPD Rate (smoothed trend)
3M Avg 30Plus DPD Rate =
```

```
AVERAGEX (
DATESINPERIOD ( DimDate\[Date\], MAX ( DimDate\[Date\] ), -3,
MONTH ),
CALCULATE ( \[30Plus DPD Rate by Month\] )
)
// 6-Month DPD Trend Direction (for conditional formatting)
DPD Trend Direction =
VAR Current3M = \[3M Avg 30Plus DPD Rate\]
VAR Prior3M =
CALCULATE (
\[3M Avg 30Plus DPD Rate\],
DATEADD ( DimDate\[Date\], -3, MONTH )
)
RETURN
SWITCH (
TRUE (),
Current3M \> Prior3M \* 1.1, \"Deteriorating\",
Current3M \< Prior3M \* 0.9, \"Improving\",
\"Stable\"
)
```

D. Roll Rate (Transition) Analysis

Roll rate analysis tracks how loans migrate between DPD buckets month-over-month. This is the most critical predictive tool for estimating future losses. Rating agencies use roll rates to calibrate their loss models, and investors monitor them for early warning signals.

Roll Rate Data Requirements:

Column Name	Data Type	Description	Example
SnapshotDate	DATE	Current reporting period	2024-10-31
LoanID	VARCHAR(20)	Loan identifier	ZAAUTO2024-1-00001
FromBucket	VARCHAR(15)	DPD bucket in prior period	Current
ToBucket	VARCHAR(15)	DPD bucket in current period	1-29 DPD
TransitionType	VARCHAR(10)	Cure/Roll/Static/Payoff/Default	Roll
Balance	DECIMAL(15,2)	Balance at transition	487,250.00

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Roll Rate Matrix - Standard Format:

The roll rate matrix shows the probability of a loan moving from one DPD bucket to another in a single month. This matrix is displayed as a heatmap in Power BI:

From ↓ / To →	Current	1-29 DPD	30-59 DPD	60-89 DPD	90+ DPD	Default
Current	96.5%	2.8%	0.5%	0.1%	0.05%	0.05%
1-29 DPD	45.0%	30.0%	20.0%	3.0%	1.5%	0.5%
30-59 DPD	15.0%	10.0%	35.0%	30.0%	8.0%	2.0%
60-89 DPD	5.0%	3.0%	7.0%	30.0%	40.0%	15.0%
90+ DPD	2.0%	1.0%	2.0%	5.0%	50.0%	40.0%

DAX for Roll Rate Calculations:

```
// Roll Rate from Bucket A to Bucket B

Roll Rate =

VAR FromBucket = SELECTEDVALUE ( RollRateMatrix\[FromBucket\]
)

VAR ToBucket = SELECTEDVALUE ( RollRateMatrix\[ToBucket\] )

VAR CurrentDate = MAX ( DimDate\[Date\] )

VAR PriorDate = EDATE ( CurrentDate, -1 )

// Loans that were in FromBucket last month

VAR LoansInFromBucket =

CALCULATETABLE (

VALUES ( FactLoanPerformance\[LoanID\] ),

FactLoanPerformance\[DPD_Bucket\] = FromBucket,

DimDate\[Date\] = PriorDate
```

```
)
VAR BalanceInFromBucket =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Bucket\] = FromBucket,
DimDate\[Date\] = PriorDate
)
// Of those, how many moved to ToBucket this month
VAR BalanceTransitioned =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Bucket\] = ToBucket,
DimDate\[Date\] = CurrentDate,
TREATAS ( LoansInFromBucket, FactLoanPerformance\[LoanID\] )
)
RETURN
DIVIDE ( BalanceTransitioned, BalanceInFromBucket )
// Cure Rate (loans returning to Current from any delinquent
bucket)
Cure Rate =
VAR CurrentDate = MAX ( DimDate\[Date\] )
VAR PriorDate = EDATE ( CurrentDate, -1 )
VAR DelinqLastMonth =
CALCULATETABLE (
VALUES ( FactLoanPerformance\[LoanID\] ),
FactLoanPerformance\[DPD_Days\] \>= 30,
DimDate\[Date\] = PriorDate
```

```
)
VAR DelinqBalanceLastMonth =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 30,
DimDate\[Date\] = PriorDate
)
VAR CuredBalance =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] = 0,
DimDate\[Date\] = CurrentDate,
TREATAS ( DelinqLastMonth, FactLoanPerformance\[LoanID\] )
)
RETURN DIVIDE ( CuredBalance, DelinqBalanceLastMonth )
// Roll-to-Default Rate (from 60-89 DPD to 90+ DPD)
Roll to Default Rate =
VAR CurrentDate = MAX ( DimDate\[Date\] )
VAR PriorDate = EDATE ( CurrentDate, -1 )
VAR Loans60to89 =
CALCULATETABLE (
VALUES ( FactLoanPerformance\[LoanID\] ),
FactLoanPerformance\[DPD_Days\] \>= 60 &&
FactLoanPerformance\[DPD_Days\] \< 90,
DimDate\[Date\] = PriorDate
)
```

```
VAR Balance60to89 =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 60 &&
FactLoanPerformance\[DPD_Days\] \< 90,
DimDate\[Date\] = PriorDate
)
VAR RolledToDefault =
CALCULATE (
SUM ( FactLoanPerformance\[CurrentBalance\] ),
FactLoanPerformance\[DPD_Days\] \>= 90,
DimDate\[Date\] = CurrentDate,
TREATAS ( Loans60to89, FactLoanPerformance\[LoanID\] )
)
RETURN DIVIDE ( RolledToDefault, Balance60to89 )
```

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E. DPD Ageing Report

The ageing report shows how long loans have been in their current delinquent state. This helps distinguish between newly delinquent loans (more likely to cure) and chronically delinquent loans (more likely to default).

```
// Average Days in Current DPD Bucket

Avg Days in Current Bucket =

AVERAGEX (

FILTER ( FactLoanPerformance, FactLoanPerformance\[DPD_Days\] \> 0 ),

FactLoanPerformance\[ConsecutiveMonthsDelinquent\] \* 30

)

// Chronic Delinquency Rate (30+ DPD for 3+ consecutive months)

Chronic Delinquency Rate =

VAR ChronicBalance =

CALCULATE (

SUM ( FactLoanPerformance\[CurrentBalance\] ),

FactLoanPerformance\[DPD_Days\] \>= 30,

FactLoanPerformance\[ConsecutiveMonthsDelinquent\] \>= 3

)

RETURN DIVIDE ( ChronicBalance, SUM (

FactLoanPerformance\[CurrentBalance\] ) )

// First-Time Delinquency Rate (new entries to 30+ DPD)

First Time 30Plus DPD Rate =

VAR CurrentDate = MAX ( DimDate\[Date\] )

VAR PriorDate = EDATE ( CurrentDate, -1 )

VAR NewlyDelinquent =

CALCULATE (
```

```
SUM ( FactLoanPerformance\[CurrentBalance\] ),

FactLoanPerformance\[DPD_Days\] \>= 30,

FactLoanPerformance\[DPD_Bucket_Prior\]   IN   {   \"Current\",
\"1-29 DPD\"

},

DimDate\[Date\] = CurrentDate

)

VAR PoolBalance = CALCULATE ( SUM (

FactLoanPerformance\[CurrentBalance\]   ),   DimDate\[Date\]   =
CurrentDate

)

RETURN DIVIDE ( NewlyDelinquent, PoolBalance )
```

F. DPD Dashboard Page Specification

The Power BI DPD dashboard page should contain the following visuals:

1. KPI Cards Row: Current %, 30+ DPD %, 60+ DPD %, 90+ DPD %, with trend arrows and conditional formatting
2. Stacked Bar Chart: DPD distribution by balance, coloured by bucket (green to red gradient)
3. Line Chart: 30+ DPD Rate monthly trend over 24 months, with 3M moving average overlay
4. Roll Rate Heatmap: Matrix visual showing month-over-month transition probabilities
5. Table: Loan-level detail of all 60+ DPD loans with borrower attributes
6. Slicer Bar: Pool, Vintage Year, Region, Vehicle Type, Origination Channel
7. Gauge Chart: Delinquency trigger level vs current level (PASS/FAIL indicator)
8. Small Multiples: DPD trend split by vintage year for comparison

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A2.9 Static Loss Data - Vintage/Static Pool Analysis

Static pool analysis (also called vintage analysis or static loss analysis) is the gold standard methodology used by rating agencies to project expected losses for a securitised pool. Unlike dynamic analysis which looks at the current snapshot, static pool analysis tracks a fixed group of loans from origination to termination, regardless of when new loans are added or existing loans pay off.

A. Static Pool Concept

Definition:

A static pool is a fixed set of loans originated during a specific time period (e.g., Q1 2022 originations). The performance of this pool is tracked cumulatively over time from origination (Month 0) onwards. No loans are added to or removed from the static pool after the cutoff. This pure measurement of credit performance eliminates the distorting effects of pool growth, seasoning mix changes, and portfolio turnover.

Why Rating Agencies Require Static Loss Data:

- Eliminates denominator effect: New originations don't dilute delinquency rates
- Pure credit performance: Measures only credit deterioration, not portfolio management
- Vintage comparison: Enables direct comparison across origination cohorts
- Loss curve projection: Allows extrapolation of partial curves to project ultimate losses
- Originator assessment: Evaluates whether underwriting quality is stable, improving, or deteriorating over time

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B. Static Pool Data Structure

The static pool data table must be constructed separately from the dynamic performance data. It requires a fixed denominator (original pool size at month 0) and cumulative numerators tracked monthly:

Column Name	Data Type	Description	Example
VintageID	VARCHAR(10)	Origination cohort	2022-Q1
VintageStartDate	DATE	First day of vintage period	2022-01-01
OriginalLoanCount	INT	Loans in static pool at Month 0	1,250
OriginalPoolBalance	DECIMAL(15,2)	Total balance at Month 0	125,00,00,000
MonthsOnBook	INT	Months since pool inception	24
CumulativeDefaults_Count	INT	Total defaults since Month 0	37
CumulativeDefaults_Balance	DECIMAL(15,2)	Default balance since Month 0	3,42,50,000
CumulativeGrossLoss	DECIMAL(15,2)	Gross loss amount since Month 0	2,10,00,000
CumulativeRecoveries	DECIMAL(15,2)	Recoveries on defaults since Month 0	65,00,000
CumulativeNetLoss	DECIMAL(15,2)	Net loss (Gross - Recoveries)	1,45,00,000
CumulativeNetLossRate	DECIMAL(8,6)	Net Loss / Original Pool Balance	0.011600

CumulativePrepayments	DECIMAL(15,2)	Voluntary prepayments since Month 0	18,75,00,000
RemainingPoolBalance	DECIMAL(15,2)	Outstanding balance of surviving loans	87,50,00,000
PoolFactor	DECIMAL(5,4)	Remaining / Original balance	0.7000
CurrentDelinq30Plus	DECIMAL(8,6)	30+ DPD rate of surviving loans	0.035000

C. DAX for Static Pool / Vintage Analysis

```
// Cumulative Net Loss Rate by Vintage

Static CNL Rate =

VAR SelectedVintage = SELECTEDVALUE ( StaticPool\[VintageID\] )

VAR MaxMOB = MAX ( StaticPool\[MonthsOnBook\] )

VAR CumNetLoss =

CALCULATE (

MAX ( StaticPool\[CumulativeNetLoss\] ),

StaticPool\[VintageID\] = SelectedVintage,

StaticPool\[MonthsOnBook\] = MaxMOB

)

VAR OrigBalance =

CALCULATE (

MAX ( StaticPool\[OriginalPoolBalance\] ),

StaticPool\[VintageID\] = SelectedVintage

)

RETURN DIVIDE ( CumNetLoss, OrigBalance )
```

```
// Projected Ultimate CNL (Extrapolation using loss curve)
Projected Ultimate CNL =
VAR CurrentCNL = \[Static CNL Rate\]
VAR CurrentMOB = MAX ( StaticPool\[MonthsOnBook\] )
VAR ExpectedLifeMonths = 60 // 5-year avg life for auto loans
// Use historical loss timing curve to extrapolate
VAR LossCurveCompletion =
CALCULATE (
MAX ( LossTimingCurve\[CumulativePctOfLoss\] ),
LossTimingCurve\[MonthsOnBook\] = CurrentMOB
)
RETURN DIVIDE ( CurrentCNL, LossCurveCompletion )
// Vintage Comparison - CNL at Same Seasoning Point
CNL at 12 Months =
CALCULATE (
MAX ( StaticPool\[CumulativeNetLossRate\] ),
StaticPool\[MonthsOnBook\] = 12
)
CNL at 24 Months =
CALCULATE (
MAX ( StaticPool\[CumulativeNetLossRate\] ),
StaticPool\[MonthsOnBook\] = 24
)
// Marginal Loss Rate (loss in current month only)
Marginal Monthly Loss Rate =
VAR CurrentMOB = MAX ( StaticPool\[MonthsOnBook\] )
```

```
VAR CurrentCNL = \[Static CNL Rate\]

VAR PriorCNL =

CALCULATE (

\[Static CNL Rate\],

StaticPool\[MonthsOnBook\] = CurrentMOB - 1

)

RETURN CurrentCNL - PriorCNL

// Loss Curve Slope (rate of change - early warning)

Loss Curve Acceleration =

VAR CurrentMarginal = \[Marginal Monthly Loss Rate\]

VAR PriorMarginal =

CALCULATE (

\[Marginal Monthly Loss Rate\],

StaticPool\[MonthsOnBook\] = MAX(StaticPool\[MonthsOnBook\]) -
1

)

RETURN

IF (

CurrentMarginal \> PriorMarginal \* 1.2, \"Accelerating\",

IF    (    CurrentMarginal    \<    PriorMarginal    \*    0.8,
\"Decelerating\",

\"Stable\" )

)
```

D. Loss Timing Curve (S&P/Moody's Requirement)

The loss timing curve defines what percentage of total expected lifetime losses occur in each month. Rating agencies use this to time-stress cash flows. The standard loss timing curve for Indian auto loans peaks around months 18-30:

MOB Range	% of Total Loss	Cumulative %	Typical Auto	Typical RMBS	Typical Cards	Typical CLO
0-6	5%	5%	Low	Very Low	Moderate	Very Low
7-12	12%	17%	Rising	Low	Moderate	Low
13-18	18%	35%	Peak	Rising	Moderate	Rising
19-24	22%	57%	Peak	Rising	Moderate	Moderate
25-36	25%	82%	Declining	Peak	Moderate	Peak
37-48	12%	94%	Low	Declining	Declining	Declining
49-60	6%	100%	Tail	Declining	Low	Tail

A2.10 Dynamic Loss Data - Point-In-Time Portfolio Analysis

Dynamic loss data measures the performance of the current pool at a specific point in time. Unlike static pool analysis, the denominator (pool size) changes every month as loans prepay, amortise, default, or are added. Dynamic analysis is used for monthly investor reporting, trigger testing, and operational portfolio management.

A. Key Differences: Static vs Dynamic

Dimension	Static Loss Data	Dynamic Loss Data
Pool Definition	Fixed cohort at origination	Current outstanding pool
Denominator	Original pool balance (fixed)	Current pool balance (changes monthly)
Time Horizon	From origination to final payment	Current reporting period
New Originations	Excluded after cutoff	Included in pool
Prepayments	Tracked but pool stays fixed	Reduce the current denominator
Primary Use	Rating agency loss projection	Monthly investor reporting
Key Metric	Cumulative Net Loss Rate	Annualised Default/Loss Rate
Distortion Risk	None (pure measurement)	Denominator effect from pool changes
DAX Complexity	Moderate (time-series)	High (point-in-time with prior period references)

B. Dynamic Loss Data Structure

Column Name	Data Type	Description	Example
ReportingDate	DATE	End of reporting period	2024-10-31
BOP_LoanCount	INT	Loan count at start of month	475
BOP_Balance	DECIMAL(15,2)	Pool balance at start of month	32,50,00,000

NewDefaults_Count	INT	Loans defaulting this month	3
NewDefaults_Balance	DECIMAL(15,2)	Balance of new defaults	18,50,000
GrossLoss_ThisMonth	DECIMAL(15,2)	Gross losses realised this month	12,00,000
Recoveries_ThisMonth	DECIMAL(15,2)	Recoveries this month	3,50,000
NetLoss_ThisMonth	DECIMAL(15,2)	Net loss this month	8,50,000
Prepayments_ThisMonth	DECIMAL(15,2)	Voluntary prepayments	1,25,00,000
ScheduledAmort	DECIMAL(15,2)	Scheduled principal repayment	85,00,000
EOP_Balance	DECIMAL(15,2)	Pool balance at end of month	30,22,00,000
EOP_LoanCount	INT	Loan count at end of month	468
CollectionsTotal	DECIMAL(15,2)	Total cash collected	2,85,00,000
CollectionEfficiency	DECIMAL(5,4)	Collections / Billing amount	0.9720

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C. Dynamic Loss DAX Measures

Measure Group 1: Period Loss Rates

```
// Monthly Default Rate (MDR)
Monthly Default Rate =
VAR BOP_Balance =
CALCULATE (
SUM ( DynamicLoss\[BOP_Balance\] ),
DimDate\[Date\] = MAX ( DimDate\[Date\] )
)
VAR NewDefaults =
CALCULATE (
SUM ( DynamicLoss\[NewDefaults_Balance\] ),
DimDate\[Date\] = MAX ( DimDate\[Date\] )
)
RETURN DIVIDE ( NewDefaults, BOP_Balance )
// Annualised Default Rate (Constant Default Rate - CDR)
CDR = 1 - POWER ( 1 - \[Monthly Default Rate\], 12 )
// Monthly Loss Rate
Monthly Net Loss Rate =
VAR AvgBalance =
DIVIDE (
CALCULATE ( SUM ( DynamicLoss\[BOP_Balance\] ) ) +
CALCULATE ( SUM ( DynamicLoss\[EOP_Balance\] ) ),
2
)
RETURN DIVIDE (
```

```
CALCULATE ( SUM ( DynamicLoss\[NetLoss_ThisMonth\] ) ),

AvgBalance

)

// Annualised Net Loss Rate

Annualised Net Loss Rate = \[Monthly Net Loss Rate\] \* 12

// Rolling 12-Month Net Loss Rate

Rolling 12M Net Loss Rate =

VAR Last12MLoss =

CALCULATE (

SUM ( DynamicLoss\[NetLoss_ThisMonth\] ),

DATESINPERIOD ( DimDate\[Date\], MAX ( DimDate\[Date\] ), -12,
MONTH )

)

VAR AvgPoolBalance =

AVERAGEX (

DATESINPERIOD ( DimDate\[Date\], MAX ( DimDate\[Date\] ), -12,
MONTH

),

CALCULATE ( SUM ( DynamicLoss\[BOP_Balance\] ) )

)

RETURN DIVIDE ( Last12MLoss, AvgPoolBalance )
```

Measure Group 2: Prepayment Speed

```
// Single Month Mortality (SMM)

SMM =

VAR BOP = CALCULATE ( SUM ( DynamicLoss\[BOP_Balance\] ) )

VAR SchedPrin = CALCULATE ( SUM ( DynamicLoss\[ScheduledAmort\] ) )

VAR Prepay = CALCULATE ( SUM ( DynamicLoss\[Prepayments_ThisMonth\] )

)

RETURN DIVIDE ( Prepay, BOP - SchedPrin )

// Conditional Prepayment Rate (CPR - Annualised)

CPR = 1 - POWER ( 1 - \[SMM\], 12 )

// Absolute Prepayment Speed (ABS) - Alternative Measure

ABS Prepayment Speed =

DIVIDE (

CALCULATE ( SUM ( DynamicLoss\[Prepayments_ThisMonth\] ) ),

CALCULATE ( SUM ( DynamicLoss\[BOP_Balance\] ) )

)
```

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Measure Group 3: Recovery and Severity

```
// Recovery Rate
Recovery Rate =
DIVIDE (
CALCULATE ( SUM ( DynamicLoss\[Recoveries_ThisMonth\] ) ),
CALCULATE ( SUM ( DynamicLoss\[GrossLoss_ThisMonth\] ) )
)
// Loss Severity (1 - Recovery Rate)
Loss Severity = 1 - \[Recovery Rate\]
// Average Recovery Lag (months from default to recovery)
Avg Recovery Lag =
AVERAGEX (
FILTER (
FactCreditEvents,
FactCreditEvents\[RecoveryDate\] \<\> BLANK ()
),
DATEDIFF (
FactCreditEvents\[DefaultDate\],
FactCreditEvents\[RecoveryDate\],
MONTH
)
)
// Cumulative Recovery Rate (all-time)
Cumulative Recovery Rate =
DIVIDE (
CALCULATE ( SUM ( DynamicLoss\[Recoveries_ThisMonth\] ), ALL (
DimDate
```

```
) ),
CALCULATE ( SUM ( DynamicLoss\[GrossLoss_ThisMonth\] ), ALL (
DimDate
) )
)
```

Measure Group 4: Cash Flow Analysis

```
// Excess Spread
Monthly Excess Spread =
VAR InterestCollected = \[Pool Interest Collections\]
VAR InvestorCouponDue = \[Total Tranche Interest Due\]
VAR ServicingFee = \[Servicer Fee Amount\]
VAR TrusteeFee = \[Trustee Fee Amount\]
VAR LossesAbsorbed = CALCULATE ( SUM (
DynamicLoss\[NetLoss_ThisMonth\] ) )
RETURN InterestCollected - InvestorCouponDue - ServicingFee -
TrusteeFee - LossesAbsorbed
// Annualised Excess Spread Rate
Annualised Excess Spread =
DIVIDE ( \[Monthly Excess Spread\], CALCULATE ( SUM (
DynamicLoss\[BOP_Balance\] ) ) ) \* 12
// Months of Excess Spread Reserve
Excess Spread Coverage Months =
VAR MonthlyLoss = CALCULATE ( SUM (
DynamicLoss\[NetLoss_ThisMonth\] )
)
RETURN
IF ( MonthlyLoss \> 0,
```

```
DIVIDE ( \[Monthly Excess Spread\], MonthlyLoss ),
999 // No losses
)
```

D. Dynamic Loss Dashboard Page Specification

The Dynamic Loss Analysis dashboard page should contain:

1. KPI Cards: CDR, CPR, Recovery Rate, Annualised Net Loss Rate, Excess Spread
2. Waterfall Chart: BOP Balance → Scheduled Amort → Prepayments → Defaults → EOP Balance
3. Line Chart: Rolling 12M Net Loss Rate and CDR trend over 24 months
4. Stacked Area Chart: Monthly collections breakdown (Interest vs Principal vs Prepayments vs Recoveries)
5. Scatter Plot: CDR vs CPR by vintage (bubble size = pool balance)
6. Table: Monthly performance summary (12-month history)
7. Gauge: Excess spread level vs trigger thresholds

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E. Reconciliation: Static vs Dynamic Loss Measures

The DAX model must include reconciliation measures to ensure static and dynamic loss data are consistent. Any variance must be explained:

```
// Static vs Dynamic Loss Reconciliation

Loss Data Reconciliation Variance =

VAR StaticCNL = \[Static CNL Rate\]

VAR DynamicCumLoss =

DIVIDE (

CALCULATE ( SUM ( DynamicLoss\[NetLoss_ThisMonth\] ), ALL ( DimDate )

),

CALCULATE ( SUM ( DynamicLoss\[BOP_Balance\] ), DimDate\[Date\] = MIN

( DimDate\[Date\] ) )

)

RETURN ABS ( StaticCNL - DynamicCumLoss )

// Variance Explanation Flag

Reconciliation Status =

IF (

\[Loss Data Reconciliation Variance\] \< 0.001,

\"Reconciled\",

\"Variance \> 10bps - Investigate\"

)
```

A2.11 Cross-Asset Class Data Comparison

The following table summarises the key differences in data requirements across securitisation asset classes. This matrix helps the data analyst understand which DAX measures and dashboard pages are needed for each asset class:

Data Element	Auto ABS	RMBS	CLO	Credit Card ABS
Primary Collateral	Vehicle	Property	Corporate Loans	Receivables
Amortisation	Fully amortising	Mostly amortising	Bullet/Amortising	Revolving
Average Life	2-4 years	5-10 years	4-7 years	Controlled amort
Key Risk Metric	Default Rate	LTV/DSCR	WARF/Leverage	Payment Rate
Prepayment Measure	ABS/CPR	CPR/PSA	Repricing risk	N/A (revolving)
Recovery Driver	Vehicle resale	Property sale	Distressed debt	Write-off
Typical Recovery	40-60%	50-80%	40-70%	0-10%
AAA Stress Multiple	3.0-4.0x	2.5-3.5x	Varies by WARF	4.0-5.0x
Key Concentration	Geographic	Geographic	Industry	Vintage/FICO
Compliance Tests	OC/IC	OC/IC/DSC	OC/IC/WARF/WAS	Excess spread
IFRS 9 Approach	General model	General model	General model	General model

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A2.12 Complete Dax Measure Library For Investor Reporting

The following DAX measures form the complete measure library needed to build an investor-grade securitisation dashboard. Each measure is production-ready and has been performance-optimised.

A. Pool Health Measures

```
// 1. Pool Factor
Pool Factor = DIVIDE(SUM(Loans\[CurrentBalance\]),
CALCULATE(SUM(Loans\[OriginalLoanAmount\]),ALL(DimDate)))
// 2. Delinquency Rate (by balance)
Delinquency Rate =
DIVIDE(
CALCULATE(SUM(Loans\[CurrentBalance\]),
Loans\[DelinquencyDays\]\>=30),
SUM(Loans\[CurrentBalance\])
)
// 3. Serious Delinquency Rate (60+ DPD)
Serious Delinquency Rate =
DIVIDE(
CALCULATE(SUM(Loans\[CurrentBalance\]),
Loans\[DelinquencyDays\]\>=60),
SUM(Loans\[CurrentBalance\])
)
// 4. Cumulative Default Rate
Cumulative Default Rate =
VAR CumDefaults = CALCULATE(SUM(Loans\[CurrentBalance\]),
Loans\[IsDefaulted\]=TRUE, ALL(DimDate))
VAR OriginalPool = CALCULATE(SUM(Loans\[OriginalLoanAmount\]),
```

```
ALL(DimDate))

RETURN DIVIDE(CumDefaults, OriginalPool)

// 5. Collection Efficiency

Collection Efficiency =

DIVIDE(SUM(Loans\[ActualPayment\]),
SUM(Loans\[ScheduledPaymentDue\]))

// 6. Constant Default Rate (CDR) - Annualised

CDR =

VAR MonthlyDefault = \[Defaults This Month\]

VAR BOP_Balance = CALCULATE(SUM(Loans\[CurrentBalance\]),

PREVIOUSMONTH(DimDate\[Date\]))

VAR MDR = DIVIDE(MonthlyDefault, BOP_Balance)

RETURN 1 - POWER(1 - MDR, 12)
```

B. Vintage Analysis Measures

```
// 7. Vintage Cumulative Loss Curve

Vintage Cumulative Loss =

VAR                      SelectedVintage                      =
SELECTEDVALUE(DimLoan\[OriginationYear\])

VAR MaxMOB = MAX(FactLoanPerformance\[MonthsOnBook\])

RETURN

CALCULATE(

SUM(FactCreditEvents\[NetLoss\]),

DimLoan\[OriginationYear\] = SelectedVintage,

FactLoanPerformance\[MonthsOnBook\] \<= MaxMOB,

ALL(DimDate)

)
```

C. Credit Enhancement Tracking

```
// 8. Credit Enhancement Percentage

Credit Enhancement % =

VAR SubordinationAmt = \[Subordinated Tranche Balance\]

VAR ExcessSpreadReserve = \[Reserve Account Balance\]

VAR OC_Amount = SUM(Loans\[CurrentBalance\]) - \[Senior Tranche

Balance\]

RETURN

DIVIDE(SubordinationAmt + ExcessSpreadReserve + OC_Amount,

SUM(Loans\[CurrentBalance\]))

// 9. Reserve Account Adequacy

Reserve Adequacy =

DIVIDE(\[Reserve Account Balance\], \[3M Avg Monthly Losses\])
```

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PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

Note on the Simulation Concept: The gamified simulation concept presented in this section is provided as a reference framework. Students may choose to follow this concept or design their own original gamified simulation - with a different name, structure, narrative, or game mechanics - as long as it meets the assessment criteria outlined in Part F.

SECTION B1: SECURITISATION RISK ARENA - SIMULATION OVERVIEW

B1.1 Business Rationale And Problem Statement

Securitisation Risk Arena is a browser-based gamified simulation platform designed to train and assess data analysts in the specialised discipline of securitisation risk assessment through realistic, time-pressured, scenario-driven challenges using advanced DAX analytics. The player assumes the role of Lead Data Analyst at a fictional global bank - Meridian Structured Finance - managing a $50 billion portfolio of securitised assets across MBS, ABS, CLO, and CMBS structures. Each session simulates a full business day in the life of a securitisation data analyst: from receiving the morning portfolio health report to delivering risk dashboards to the Chief Risk Officer by end of day.

The Training Gap This Platform Solves:

Traditional securitisation analytics training relies on passive reading of rating agency methodologies and regulatory guidelines, occasional shadowing of senior analysts during deal closings or portfolio reviews, and learning by making errors on live reporting systems. The consequences of errors in securitisation analytics are particularly severe: an incorrect ECL computation can misstate the bank's provision requirements by millions; a flawed waterfall calculation can misallocate cash flows between senior and subordinated tranches; a missed delinquency trend can delay early-warning triggers, leading to rating downgrades and investor losses.

India's securitisation market has grown significantly following the RBI's revised securitisation framework (September 2021), with issuance volumes exceeding ₹1.8 lakh crore annually. Global structured finance markets process trillions in outstanding securities. Yet there is no standardised, scalable training platform for securitisation data analysts that combines DAX proficiency with domain expertise. Securitisation Risk Arena creates a zero-risk environment where learners experience every possible analytics scenario - from routine monthly reporting to crisis-mode stress testing - before encountering them in a live portfolio.

Target Audience:

- Primary: Undergraduate and MBA students preparing for data analytics, structured finance, and risk management roles at banks, NBFCs, and rating agencies.
- Secondary: Working professionals in credit risk, portfolio analytics, and investor reporting seeking to upskill on advanced DAX techniques for securitisation.
- Tertiary: Bank HR and training teams seeking to assess structured finance analytics competency during hiring.

B1.2 Simulation Architecture

Core Components:

- Portfolio Library: Pre-built datasets for 6 fictional securitisation pools (auto loan ABS, RMBS, CLO, credit card ABS, CMBS, student loan ABS) with 500+ loans per pool, realistic field structures matching rating agency requirements.
- Market Simulator: Monthly data generator that simulates delinquency transitions, prepayment behaviour, loss severity, and recovery rates based on parameterised economic scenarios - benign, moderate stress, severe stress, and crisis.
- DAX Challenge Engine: Presents the player with specific analytical tasks requiring DAX measure construction - from basic aggregations to complex iterator patterns, virtual table computations, and time-intelligence measures.
- Waterfall Engine: Simulated cash flow waterfall that allocates collections across tranches based on player-built DAX measures; errors in the waterfall formula cascade through tranche payments, revealing mistakes immediately.
- Rating Agency Module: Requires the player to produce loan-level data tapes, pool stratification tables, and performance metrics matching S&P, Moody's, Fitch, CRISIL, and ICRA specifications.
- ECL Engine: Embedded IFRS 9 computation module that evaluates player-built PD, LGD, and EAD measures for stage classification and expected credit loss computation.
- Scoring Engine: Awards points for DAX accuracy, query performance, analytical completeness, and business insight; deducts points for errors, slow queries, and missed risk signals.
- Learning Layer: Every error triggers an explanation pop-up with the correct DAX pattern, optimisation suggestion, and reference to the relevant securitisation concept.

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B1.3 Game Modes

Campaign Mode (Primary Mode):

8 progressive levels, each spanning 3 simulated business days:

- Level 1 - Data Foundation: Single auto loan pool; basic CALCULATE and SUM patterns; simple delinquency bucket counts. Focus: filter context, basic aggregation, and data model understanding.
- Level 2 - Time Intelligence: Same pool with 12 months of history; YTD, QTD, MTD, rolling averages, and period-over-period comparisons. Focus: time intelligence functions and date table relationships.
- Level 3 - Iterator Mastery: Two pools (auto ABS + RMBS); weighted averages using SUMX, AVERAGEX; concentration measures using TOPN and RANKX. Focus: row context, iterator functions, and context transition.
- Level 4 - Virtual Tables: Three pools; SUMMARIZE, ADDCOLUMNS, GENERATE patterns; dynamic stratification tables; pool-level summary computations. Focus: virtual table construction and advanced DAX patterns.
- Level 5 - Waterfall Wizard: Complete cash flow waterfall allocation across senior, mezzanine, and equity tranches; trigger and covenant testing; credit enhancement tracking. Focus: sequential allocation logic in DAX.
- Level 6 - IFRS 9 ECL: Full ECL computation framework; PD term structures, LGD estimation, EAD computation; stage classification and migration; forward-looking macroeconomic overlay. Focus: regulatory capital computation.
- Level 7 - Rating Agency Reporting: Complete loan-level data tape generation for S&P, Moody's, and CRISIL; all required fields populated; validation checks passed. Focus: investor reporting and regulatory compliance.
- Level 8 - Crisis Mode: Market stress scenario with spiking delinquencies, plummeting recovery rates, and trigger breaches across all 6 pools simultaneously. Fund manager demands real-time dashboards; rating agency requests emergency data tape; board wants stress testing results by EOD.

Practice Mode:

- Free-play access to any single scenario type - player selects pool, asset class, scenario difficulty, and DAX complexity level.
- Ideal for practising a specific DAX pattern (e.g., CALCULATETABLE, nested iterators, virtual table joins).
- No penalties; unlimited attempts; step-by-step DAX guidance available.

Assessment Mode:

- Timed, scored, and monitored simulation session.
- Used for formal assessment during internship evaluation or bank hiring process.
- No hints; no guidance; full scoring rubric applied.

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SECTION B2: SCORING SYSTEM AND GAME MECHANICS

B2.1 1000-Point Scoring Framework

Scoring Category	Points	Key Criteria
DAX Accuracy & Complexity	300	All measures correct across test scenarios; appropriate patterns used; proper context transition
Query Performance	150	All measures <100ms in DAX Studio; efficient use of variables and filter propagation
Waterfall & Tranche Allocation	200	Complete waterfall with correct sequential allocation, trigger testing, and credit enhancement tracking
IFRS 9 ECL Computation	150	ECL matches Excel crosscheck within 0.01% variance; correct stage classification
Investor Reporting Completeness	100	All 6 monthly servicer report sections automated with correct DAX measures
Dashboard Quality & Business Insight	100	8+ interactive pages; drill-through; RLS for 5+ roles; business insights narrative
TOTAL	1000	

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Penalties:

- Hard-coded value in a DAX measure (instead of dynamic calculation): −50 points per instance.
- Measure that returns incorrect sign (gain shown as loss or vice versa): −40 points.
- Missing delinquency trend that should have triggered early warning: −30 points.
- Waterfall misallocation affecting tranche payments: −25 points per tranche.
- ECL stage classification error: −20 points per misclassified segment.

B2.2 Progression System

Player Levels (8 Levels):

Level	Title	XP Required	Unlocks
1	Data Analyst Trainee	0	Tutorial scenarios; single pool; full DAX guidance
2	Junior Data Analyst	500	Campaign Levels 1–2; basic DAX and time intelligence
3	Data Analyst	1,500	Campaign Levels 1–4; iterator functions and virtual tables
4	Senior Data Analyst	3,500	Campaign Levels 1–6; waterfall calculations and ECL models
5	Lead Analyst	7,000	All Campaign Levels; complete investor reporting suite
6	VP Structured Finance Analytics	15,000	Crisis scenarios; real-time dashboard construction under pressure
7	Director of Securitisation Analytics	25,000	All scenarios mastered; can design challenge sets for others

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8	Chief Risk Analytics Officer	40,000	Teach mode enabled; enterprise deployment expertise

Achievement Badges (25 Badges):

Dax Mastery Badges:

- DAX Master: Write 50+ production-grade measures across all complexity levels.
- Context Ninja: Implement 10+ CALCULATE with complex filter modifications without error.
- Virtual Table Architect: Build 5+ measures using SUMMARIZE/ADDCOLUMNS/GENERATE patterns.
- Performance Hawk: All measures execute in <100ms in DAX Studio.

Securitisation Expertise Badges:

- Waterfall Wizard: Implement complete tranche waterfall allocation with zero misallocation.
- ECL Engineer: Build full IFRS 9 ECL framework matching Excel crosscheck within 0.01%.
- Rating Agency Reporter: Generate complete loan-level data tapes for 3+ rating agencies.
- Vintage Veteran: Complete 5 vintage/static pool analyses with correct cumulative loss curves.

Crisis Management Badges:

- Crisis Survivor: Complete Level 8 crisis scenario with Distinction grade.
- Early Warning Champion: Detect 3 portfolio deterioration signals before trigger breach.
- Stress Test Master: Run 4+ what-if scenarios with correct ECL impact computation.

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SECTION B3: SCENARIO ENGINE DESIGN

B3.1 Scenario Categories

Category	% of Challenges	Description
Routine Scenarios	30%	Standard monthly reporting; benign markets; baseline competency and speed building
Portfolio Stress	25%	Economic deterioration; delinquency migration; stress-adjusted ECL; waterfall recomputation
Rating Agency Review	20%	Complete data tape generation; pool stratification; validation and reconciliation
Regulatory Scenarios	15%	RBI/SEBI audit simulation; methodology documentation; model validation evidence
Crisis Scenarios	10%	2008-style crisis; cascading defaults; multiple trigger breaches; extreme time pressure

B3.2 Sample Scenario: The Vintage Crisis

It is the 15th of the month - monthly reporting day. You receive the latest performance data for your auto loan ABS pool. The 2022 vintage is showing alarming deterioration: 60+ DPD has jumped from 2.1% to 4.8% in a single month, the highest roll rate from 30 DPD to 60 DPD in the pool's history. The rating agency has scheduled an emergency surveillance call for tomorrow morning and needs updated static pool loss curves, dynamic loss rates, and a revised ECL estimate by 8 AM.

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Challenge Points:

- Build DAX measures to compute the roll rate matrix showing transition probabilities for every DPD bucket for the last 6 months.
- Update the static pool vintage analysis showing cumulative loss curves for each origination quarter, highlighting the 2022 vintage divergence.
- Compute revised ECL using updated PD term structures reflecting the observed deterioration.
- Identify whether any waterfall trigger (delinquency trigger, loss trigger, or overcollateralisation trigger) has been breached or is at risk.

- Prepare a 3-page dashboard for the rating agency call: Pool Summary, Vintage Comparison, and Trigger Status.
- All DAX measures must execute in <100ms - the rating agency analyst will be navigating the dashboard live during the call.

B3.3 Business Model And Monetisation

Revenue Streams:

- B2B SaaS - Bank Structured Finance Training: Annual subscription per user; pricing range ₹12,000–20,000 per user per year; target: 50+ banks and NBFCs with structured finance teams averaging 15 analysts each.
- NISM/CFA Preparation Module: Stand-alone retail module aligned to CFA structured products and NISM certification syllabi; retail price ₹4,999 per candidate.
- Corporate Assessment Product: White-label version for bank recruitment and talent assessment; price per hiring cycle ₹75,000 per campaign.
- University Licensing: Annual license for MBA Finance, M.Com, and CA programs; ₹3–5 lakh per institution per year.
- Zetheta Securitisation Analytics Certification: Formal certification upon completing all 8 levels; fee ₹7,999 per candidate.

5-Year Revenue Projection (Base Case):

Year	B2B SaaS	Retail	University	Certification	Total
Year 1	5 banks × 10 users × ₹12K = ₹6L	200 × ₹5K = ₹10L	-	-	≈₹16 lakh
Year 2	15 banks	500 retail	3 licenses	-	≈₹1.8 crore
Year 3	30 banks	2,000 retail	10 licenses	Launch	≈₹5.2 crore
Year 4	45 banks	5,000 retail	20 licenses	Active	≈₹12 crore
Year 5	60+ institutions	10,000 retail	35 licenses	Full scale	≈₹22 crore

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

SECTION C1: THE 2008 GLOBAL FINANCIAL CRISIS - SECURITISATION MELTDOWN

C1.1 Background

The 2007–2009 Global Financial Crisis was, at its core, a securitisation crisis. US subprime mortgage-backed securities, which had been rated AAA by major rating agencies, experienced catastrophic default rates as the housing bubble burst. The crisis revealed fundamental flaws in securitisation analytics: models assumed perpetually rising house prices, correlation assumptions between mortgage defaults were grossly underestimated, and loan-level data quality was compromised by originate-to-distribute incentive misalignment. The collapse of Lehman Brothers, the bailout of AIG, and the freeze of global credit markets were direct consequences of securitisation risk being systematically underestimated.

C1.2 Data Analytics Lessons For Securitisation

Lesson	Analytics Failure	DAX Solution
Vintage Analysis	Performance not tracked by origination cohort; deterioration diluted in pool averages	CALCULATE with FILTER on origination date; DATESYTD for cumulative loss curves
Roll Rate Early Warning	Transition from 30 DPD to 60 DPD not monitored systematically	CALCULATETABLE with PREVIOUSMONTH; automated roll rate matrix and alerts
Concentration Risk	Geographic concentration in CA, FL, NV, AZ amplified losses	HHI measure using SUMX; TOPN exposure reporting; geographic heatmaps
Stress Testing Severity	Tests used mild 5–10% decline; actual exceeded 30%	What-if parameters in Power BI; multiple severity scenario measures

C1.3 Securitisation Risk Arena Simulation Scenario

Level 8 of Securitisation Risk Arena replicates a 2008-style crisis. The player must: detect vintage deterioration signals using static pool analysis, build roll rate early-warning measures, compute stress-adjusted ECL under severe scenarios, identify trigger breaches across all 6 pools, and prepare a comprehensive crisis dashboard for the Chief Risk Officer within a 60-minute time limit.

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SECTION C2: THE IL&FS CRISIS (2018) - INDIAN STRUCTURED FINANCE COLLAPSE

C2.1 Background

In September 2018, Infrastructure Leasing & Financial Services (IL&FS), one of India's largest infrastructure finance companies, defaulted on its commercial paper and bond obligations, triggering a crisis across India's structured finance and NBFC sectors. IL&FS group entities had issued securitised instruments backed by infrastructure project receivables and loan portfolios. The crisis exposed cascading risks: SPV cash flows were disrupted as underlying project payments stalled, investors in pass-through certificates (PTCs) faced payment delays, and the broader NBFC sector experienced a liquidity crunch as mutual funds and banks pulled back from structured finance exposure.

C2.2 Data Analytics Implications

Implication 1: Counterparty Concentration In Securitisation Pools

Several securitisation pools had exposure to IL&FS group entities as borrowers or guarantors. The lesson: DAX measures for obligor concentration (top 10 borrower exposure, single-name concentration limits) are critical for securitisation analytics. Power BI dashboards must highlight counterparty risk at both loan-level and pool-level.

Implication 2: Real-Time Collection Monitoring

The IL&FS crisis demonstrated that monthly reporting cycles are insufficient for detecting cash flow disruptions in securitisation pools. DAX time-intelligence measures with daily or weekly granularity, combined with collection efficiency ratios (actual collections vs scheduled collections), provide the real-time monitoring that was absent.

Implication 3: Rbi Regulatory Response

Post-IL&FS, the RBI revised its securitisation framework (September 2021), introducing stricter Minimum Holding Period and Minimum Retention Requirements. The revised framework requires originators to retain a minimum 10% of each tranche issued, ensuring skin-in-the-game. DAX measures must track compliance with these requirements in investor reporting dashboards.

C2.3 Securitisation Risk Arena Simulation Scenario

A mid-level scenario simulates the discovery that a major borrower in the auto loan ABS pool has been flagged as a stressed entity by the Reserve Bank. The player must: identify all loans linked to the entity across pools, compute the exposure as a percentage of total pool balance, assess the ECL impact under an immediate default assumption, check whether the pool's credit enhancement is sufficient to absorb the concentrated loss, and prepare a risk assessment note for the credit committee.

SECTION C3: RBI REVISED SECURITISATION FRAMEWORK (2021)

C3.1 Background

The Reserve Bank of India issued its revised framework for securitisation of standard assets in September 2021 (RBI/2021-22/85 DOR.STR.REC.51/21.04.177/2021-22), replacing the 2006 guidelines. The revised framework introduced significant changes: Minimum Retention Requirement (MRR) of 10% of each tranche, Minimum Holding Period (MHP) based on repayment frequency, simplified securitisation structures with only two categories (securitisation and direct assignment), and enhanced disclosure requirements for originators and investors.

Provision	Requirement	DAX Analytics Impact
Minimum Retention (MRR)	10% of each tranche	Track retained portion per tranche; compliance dashboard
Minimum Holding Period	Based on repayment frequency	Date-based eligibility measure using DATEDIFF
Structure Types	Securitisation or Direct Assignment only	Pool classification and reporting segregation
Disclosure Requirements	Enhanced originator and investor reporting	Automated loan-level and pool-level reporting measures
Credit Enhancement	Must be disclosed and tracked	CE computation measures; tranche subordination tracking
Skin-in-the-Game	First loss or proportional retention	Retained interest valuation and risk-weighting measures

C3.2 Impact On Securitisation Analytics

The revised framework requires enhanced data analytics capabilities: loan-level data tapes must be more granular to support investor due diligence; pool performance reporting must include standardised metrics specified by RBI; originators must demonstrate that retained risk is computed correctly; and trustees must receive monthly reports with pool-level performance summaries. DAX-based analytics platforms are ideally suited for these requirements - measures can compute MRR compliance, generate standardised pool reports, and produce loan-level data exports matching regulatory specifications.

SECTION C4: INDUSTRY BEST PRACTICES - SECURITISATION ANALYTICS EXCELLENCE

C4.1 Industry-Leading Operational Frameworks

Straight-Through Data Processing:

Leading banks and NBFCs implement automated data pipelines from loan origination systems to Power BI models. Monthly performance data flows directly from SAP/ERP systems into the star schema data model, with DAX measures auto-computing all pool metrics. Analysts review exception reports and dashboard alerts rather than manually computing monthly figures.

Automated Early-Warning Systems:

Power BI dashboards with DAX-driven alert measures monitor delinquency migration rates, prepayment speed changes, loss severity trends, and trigger proximity. Conditional formatting and data-driven alerts notify the analytics team when any metric crosses pre-defined thresholds.

Rating Agency Reporting Automation:

Instead of manually compiling loan-level data tapes for each rating agency review, leading institutions use DAX measures to maintain continuously updated data tapes that can be exported on demand. Pool stratification tables, performance summaries, and compliance metrics are always current.

C4.2 Lessons For Intern Data Analysts

- DAX Proficiency is the Core Competency: Modern securitisation analytics is built on DAX. The ability to write complex, performant measures that produce auditable results is the single most important technical skill.
- Data Quality is Non-Negotiable: The value of analytics depends entirely on data quality. Loan-level data must be complete, consistent, and reconcilable to source systems. DAX IFERROR and validation measures are essential.
- Business Context Elevates Technical Work: A DAX measure that computes a delinquency rate is technical; understanding why that rate matters for tranche investors, rating agencies, and regulators transforms the analyst from a technician to a business advisor.
- Documentation and Audit Trail: Every DAX measure must be documented in a measure dictionary with its business purpose, formula, and expected behaviour. This is non-negotiable in regulated financial services.

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PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

SECTION D1: OVERVIEW

The 15-day project is structured as a progressive deep-dive into securitisation data analytics using advanced DAX in Power BI. The intern will move from domain immersion through data modelling, DAX measure development, dashboard construction, and case study analysis to produce a comprehensive submission package. Each phase builds on the previous. Deliverables are cumulative and must be compiled into a final submission document.

The methodology uses AI-accelerated research and development - interns are encouraged to use Claude.ai, Perplexity.ai, and similar tools for research synthesis, initial DAX structuring, and concept validation. However, all DAX measures must be tested in DAX Studio for correctness and performance, all financial figures must be verified against primary sources (RBI circulars, SEBI guidelines, rating agency methodologies), and all data models must be validated with crosscheck calculations. Unverified AI output treated as final will be penalised.

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SECTION D2: DAY-BY-DAY BREAKDOWN

Day	Focus Area	Key Deliverable
1	Domain Orientation	Securitisation landscape summary with regulatory framework map
2	Data Architecture & Star Schema	Power BI model with star schema and relationships
3	Core DAX Measures (30+)	Measure dictionary; pool balance, count, and weighted average measures
4	Time Intelligence & Trends	YTD/QTD/MTD/rolling measures; trend analysis dashboard
5	Advanced DAX & Virtual Tables	Stratification measures; concentration analysis; HHI
6	Waterfall & Tranche Allocation	Waterfall model; trigger testing; credit enhancement tracking
7	DPD & Roll Rate Analysis	DPD dashboard; roll rate matrix; ageing report
8	Vintage / Static Pool Analysis	Cumulative loss curves; vintage comparison charts
9	IFRS 9 ECL Framework	ECL model; stage classification; Excel crosscheck
10	Stress Testing & What-If	Stress testing dashboard with interactive parameters
11	Investor Reporting & Data Tapes	6 servicer report sections; rating agency data tape export
12	Security, Governance & Testing	RLS for 5+ roles; DAX Studio performance report
13	Case Studies & Documentation	First-draft report; two case study analyses

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14	Validation & Polish	Validated draft; optimised model; presentation deck
15	Final Compilation & Handover	Final PDF + presentation + model + Excel + GitHub transfer

Day 1 - Domain Orientation:

Learning Goals: Build familiarity with the securitisation ecosystem and data analytics role.

- Read RBI's revised securitisation framework (September 2021) and SEBI securitisation guidelines.
- Study rating agency methodologies: S&P LEVELS model, Moody's Idealised Default Rate tables, CRISIL securitisation criteria.
- Review the provided CSV datasets: auto loan pool (500 loans), DPD snapshot history, static pool vintage data, dynamic loss monthly data.
- Map the data fields to rating agency requirements - identify which fields are present and which need to be computed via DAX.
- Initialise PRIVATE GitHub repository with README and project structure.

Deliverable: One-page summary of India's securitisation landscape with regulatory framework map and data field mapping.

Day 2 - Data Architecture And Star Schema:

Learning Goals: Design and build the foundational data model in Power BI.

- Design star schema: DimLoan, DimDate, DimGeography, DimBorrower, DimVehicle, FactMonthlyPerformance, FactStaticPool, FactDynamicLoss.
- Import all CSV files into Power BI Desktop.
- Configure relationships and role-playing dimensions (origination date vs reporting date).
- Set up DimDate with fiscal calendar alignment and month-end date logic.
- Validate data model with basic aggregation queries.

Deliverable: Power BI model file with star schema, relationships documented in a data model diagram.

Day 3 - Core Dax Measures (Foundational):

Learning Goals: Build the first 30+ measures covering basic pool metrics.

- Build pool balance measures: Total Outstanding, Current Balance, Original Balance.
- Build count measures: Active Loans, Defaulted Loans, Prepaid Loans.
- Implement CALCULATE patterns with filter modifications for delinquency buckets (Current, 30 DPD, 60 DPD, 90 DPD, 120+ DPD).
- Create weighted average measures using SUMX: WAC, WAM, WALA, WALT, Weighted Average LTV.

Deliverable: Power BI model with 30+ measures; DAX measure dictionary (Excel) documenting each measure's formula and business purpose.

Day 4 - Time Intelligence And Trend Analysis:

Learning Goals: Implement comprehensive time intelligence for historical analysis.

- Build YTD, QTD, MTD measures for all key metrics.
- Implement rolling 3-month and 12-month averages for delinquency and loss rates.
- Create period-over-period comparison measures (vs prior month, vs prior quarter, vs prior year).
- Build trend deviation alerts: flag any metric that moves more than 2 standard deviations from its 12-month average.

Deliverable: Updated Power BI model with time intelligence measures; trend analysis dashboard page.

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Day 5 - Advanced Dax And Virtual Tables:

Learning Goals: Master iterator functions, virtual tables, and context transition.

- Build virtual table measures using SUMMARIZE and ADDCOLUMNS for pool stratification (by geography, by LTV band, by vintage, by credit score band).
- Implement TOPN and RANKX for concentration analysis (top 10 borrowers, top 5 geographic concentrations).
- Create HHI (Herfindahl-Hirschman Index) concentration measure.
- Build context transition examples showing CALCULATE inside SUMX for portfolio-level ECL computation.

Deliverable: Updated model with advanced measures; pool stratification dashboard page.

Day 6 - Waterfall And Tranche Allocation:

Learning Goals: Build complete cash flow waterfall in DAX.

- Implement sequential waterfall allocation: collections flow to senior tranche first, then mezzanine, then equity.
- Build trigger testing measures: delinquency trigger, loss trigger, overcollateralisation trigger.
- Compute credit enhancement levels for each tranche.
- Model trigger breach scenarios using what-if parameters.

Deliverable: Waterfall allocation model in Power BI; trigger status dashboard page.

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Day 7 - Dpd And Roll Rate Analysis:

Learning Goals: Build comprehensive delinquency analytics.

- Using the DPD snapshot history data, build DPD distribution measures for each reporting month.
- Implement roll rate (transition) matrix showing monthly migration between DPD buckets.
- Build DPD ageing report showing how long loans have been in each delinquency bucket.
- Create a DPD trend dashboard with conditional formatting for deteriorating metrics.

Deliverable: DPD analytics dashboard page with roll rate matrix and trend charts.

Day 8 - Vintage / Static Pool Analysis:

Learning Goals: Build vintage analysis framework using static pool data.

- Using the static pool vintage data, build cumulative default rate and cumulative loss rate curves by origination vintage.
- Create vintage comparison charts showing each vintage's loss trajectory.
- Build loss timing curve measures for rating agency requirements.
- Implement vintage-level weighted average metrics (WAC, WALA by vintage).

Deliverable: Vintage analysis dashboard page; static pool loss curve charts.

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Day 9 - Ifrs 9 Ecl Framework:

Learning Goals: Build IFRS 9 Expected Credit Loss computation in DAX.

- Implement stage classification logic: Stage 1 (current/30 DPD), Stage 2 (60–90 DPD), Stage 3 (120+ DPD/default).
- Build PD term structure measures using historical transition matrices.
- Compute LGD using recovery rate data from the pool.
- Calculate EAD for each stage.
- Compute 12-month ECL (Stage 1) and lifetime ECL (Stages 2 and 3).
- Cross-validate ECL results against an Excel computation.

Deliverable: IFRS 9 ECL model in Power BI; Excel crosscheck workbook; ECL dashboard page.

Day 10 - Stress Testing And What-If Analysis:

Learning Goals: Build parameterised stress testing framework.

- Set up what-if parameters: default rate multiplier, recovery rate haircut, prepayment speed change, interest rate shock.
- Build stress-adjusted ECL measures that respond to parameter changes.
- Create scenario comparison dashboard: base case vs moderate stress vs severe stress vs crisis.
- Model the impact of stress on waterfall trigger proximity.

Deliverable: Stress testing dashboard page with interactive what-if parameters.

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Day 11 - Investor Reporting And Rating Agency Data Tapes:

Learning Goals: Build complete investor reporting suite.

- Build all 6 monthly servicer report sections in DAX: Pool Summary, Delinquency Report, Loss & Recovery Report, Prepayment Report, Waterfall Report, Trigger Status.
- Create rating agency data tape export functionality with all required fields for S&P, Moody's, and CRISIL.
- Build ESG data fields as emerging standard additions.

Deliverable: Investor reporting dashboard pages; data tape export page.

Day 12 - Security, Governance, And Testing:

Learning Goals: Implement enterprise-grade security and validate all measures.

- Implement Row-Level Security (RLS) for 5+ roles: Portfolio Manager, Risk Analyst, Senior Management, Auditor, Servicer.
- Performance test all measures in DAX Studio (<100ms target).
- Validate ECL calculations against Excel crosscheck.
- Validate waterfall allocation against manual computation.
- Complete the DAX measure dictionary with all 80+ measures documented.

Deliverable: RLS implementation; DAX Studio performance report; validation workbook.

Day 13 - Case Study Analysis And Documentation:

Learning Goals: Apply securitisation analytics lens to real-world events.

- Write 2008 Global Financial Crisis case study from a securitisation data analytics perspective (2 pages).
- Write IL&FS Crisis case study from a structured finance analytics perspective (2 pages).
- Compile all outputs from Days 1–12 into a structured draft report following the Part A–F format.
- Write Executive Summary (2 pages).

Deliverable: Complete first-draft report; two case study analyses.

Day 14 - Validation And Polish:

Learning Goals: Final quality assurance.

- Fact-check all regulatory figures against RBI and SEBI primary sources.
- Verify all DAX measures are formula-driven with no hard-coded values.
- Re-run DAX Studio performance tests; optimise any measure exceeding 100ms.
- Proofread all text. Ensure all tables and diagrams have captions and are legible.
- Prepare a 10-slide final presentation deck.

Deliverable: Validated, proofread draft; optimised Power BI model.

Day 15 - Final Compilation, Deployment, And Handover:

Learning Goals: Deploy and submit.

- Publish Power BI model to Power BI Service.
- Compile all sections into a final PDF report.
- Transfer GitHub repository ownership to @ZethetaIntern.
- Deliver final presentation.
- Upload all deliverables to the submission portal.

Deliverable: Final PDF report + 10-slide presentation + Power BI model + Excel workbooks + GitHub repository.

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SECTION D3: MANDATORY DELIVERABLES CHECKLIST

Deliverable	Weight	Format	Description
Main Report	35%	PDF/Docx	All 6 parts compiled, professional formatting, 40+ pages, original analysis
Power BI Model	25%	.pbix	Star schema, 80+ DAX measures, 8+ dashboards, RLS implemented
Excel Workbooks	15%	.xlsx	ECL crosscheck, waterfall validation, DAX measure dictionary (100%)
Final Presentation	15%	PDF/Slides	10 slides, professional design, covers all modules
GitHub Repository	10%	PRIVATE repo	All documentation committed, model versioned, ownership transferred

Submission Instructions:

- All files to be named using convention: ProjectCode_YourName_DeliverableName (e.g., 201169C_RahulShah_MainReport)
- All PDFs must be readable and not password-protected
- Supporting Excel files must be in .xlsx format
- All daily deliverables across Days 1–15 are to be submitted on **Day 15** as a consolidated final submission. Students should combine their work into **one or two PDFs and/or one Excel file** wherever possible, rather than submitting separate files for each day. Clearly label each section within the consolidated document by its respective day and topic for ease of assessment.
- Late submissions attract the penalties specified in Part F Assessment Criteria

Technology Stack

Category	Technology	Purpose
Primary Analytics	Power BI Desktop	Dashboard development, DAX authoring
DAX Debugging	DAX Studio	Query profiling, performance testing
Data Modelling	Tabular Editor	Advanced model editing, best practice rules
Data Source	Auto Loan CSV (provided)	500 loans, 50+ fields per loan
Spreadsheet	Power Pivot (Excel)	Prototype measures, cross-validation
Version Control	GitHub (PRIVATE)	Code, PBIX, documentation
AI Code Generation	Claude.ai, ChatGPT, Cursor	DAX generation and validation

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PART E: NO-CODE TOOLS & PLATFORMS GUIDE

SECTION E1: DATA AND COMPUTATION TOOLS

E1.1 Microsoft Power BI Desktop

URL: powerbi.microsoft.com | Free: Power BI Desktop is free; Pro license for sharing.

Key Uses For This Project:

- Star Schema Data Model: Import all CSV datasets and establish relationships between dimension and fact tables.
- DAX Measure Development: Write all 80+ measures directly in the model.
- Dashboard Construction: Build 8+ interactive pages with slicers, drill-through, bookmarks, and conditional formatting.
- Row-Level Security: Configure role-based data access for 5+ user roles.
- What-If Parameters: Create interactive stress testing scenarios.

E1.2 DAX Studio

URL: https://daxstudio.org/ | Free: Completely free and open-source.

Key Uses For This Project:

- Performance Testing: Measure query execution time for all DAX measures (<100ms target).
- Query Debugging: Step through complex DAX evaluations to identify context errors.
- Data Export: Export query results for Excel crosscheck validation.

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E1.3 Google Sheets/ Microsoft Excel

Purpose: Crosscheck computations and documentation.

Key Uses For This Project:

- IFRS 9 ECL Crosscheck: Build a parallel ECL computation in Excel to validate DAX results.
- Waterfall Validation: Manual waterfall allocation computation to crosscheck DAX measures.
- DAX Measure Dictionary: Document all 80+ measures with formula, business purpose, and expected output.
- Revenue Projection: 5-year financial model for business case.

Essential Excel Functions For Securitisation Analytics:

Function	Use in This Project	Example
SUMPRODUCT()	Weighted averages (WAC, WAM, WALA, LTV)	=SUMPRODUCT(Balance,Rate)/SUM(Balance)
IF() / nested IF()	Stage classification, bucket assignment, trigger testing	=IF(DPD>90,"Stage 3",IF(DPD>30,"Stage 2","Stage 1"))
MAX(0, value)	Loss severity, CE floor calculation	=MAX(0, Face-Recovery)
VLOOKUP() / INDEX-MATCH()	PD and LGD parameter lookup by segment	=INDEX(PD_Table,MATCH(Rating,Rating_Col,0))
IFERROR()	Handle missing data, expired positions	=IFERROR(Balance/Original,0)

SECTION E2: DESIGN AND DOCUMENTATION TOOLS

E2.1 Figma / Miro / Canva

URL: https://www.figma.com/ | https://miro.com/ | https://www.canva.in/

- Figma / Miro: Dashboard wireframe design; data model ER diagram; waterfall flow diagram.
- Canva: Presentation slide design; badge design visuals for gamification.
- Lucidchart / draw.io (free): ER diagram for star schema; securitisation structure flow chart.

E2.2 Notion / Google Docs

URL: https://www.notion.com/ | https://docs.google.com/ | Purpose: Report drafting and project management.

- Draft all sections from Day 1 onwards; do not leave documentation to Days 13–15.
- Use heading styles consistently for automatic table of contents generation.
- Save all versions with date-stamped file names.

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SECTION E3: RESEARCH AND REGULATORY DATA SOURCES

E3.1 RBI Website (https://rbi.org.in/)

- RBI/2021-22/85 - Master Direction on Securitisation of Standard Assets (September 2021)
- RBI IRAC Norms - Income Recognition, Asset Classification, and Provisioning
- RBI Circular on SMA Classification
- RBI Guidelines on IFRS 9/Ind AS 109 implementation

E3.2 SEBI Website (https://www.sebi.gov.in/)

- SEBI (Issue and Listing of Securitised Debt Instruments) Regulations, 2008
- SEBI guidelines on credit rating agency methodologies for structured finance
- SEBI disclosure requirements for listed securitised instruments

E3.3 Rating Agency Methodologies

- S&P Global: LEVELS model for auto ABS, RMBS criteria, CLO criteria - spglobal.com/ratings
- Moody's: Idealised Default Rate tables, Indian ABS methodology - moodys.com
- CRISIL: Indian securitisation criteria, default and loss studies - crisil.com
- ICRA: Indian ABS and RMBS rating criteria - icra.in

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E3.4 AI Research Tools

CLAUDE.AI: Use for explaining DAX patterns with securitisation context; summarising rating agency methodologies; drafting initial DAX measures; brainstorming scenario design. Always verify DAX measures in DAX Studio. Claude is a research accelerator, not a primary source for regulatory compliance.

Example Prompts For This Project:

- 'Write a DAX measure for cumulative default rate by vintage quarter using CALCULATETABLE and DATESYTD.'
- 'Explain the S&P LEVELS model for auto ABS with the key stress scenarios.'
- 'Help me design a roll rate matrix DAX measure for monthly transition probabilities.'
- 'Build an IFRS 9 stage classification DAX measure using SWITCH and CALCULATE.'

E3.5 DAX Learning Resources

- DAX Guide (https://dax.guide/): Comprehensive reference for all DAX functions.
- SQLBI (https://www.sqlbi.com/): Advanced DAX patterns by Marco Russo and Alberto Ferrari.
- DAX Studio documentation: Performance profiling and query optimisation.
- Tabular Editor (https://tabulareditor.com/): Advanced model editing and best practice analyser.

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PART F: SUBMISSION PROTOCOL

Projects are assessed using a comprehensive 1000-point rubric covering seven dimensions: Problem Understanding, Solution Quality, Research & Analysis, Presentation & Clarity, Innovation & Creativity, Feasibility & Practicality and CV alignment. Assessment is conducted through a combination of manual review and AI-powered evaluation.

SECTION F1: AUTOMATIC DISQUALIFIERS

The following issues result in automatic project rejection requiring re-submission:

- Repository not transferred to ZethetaIntern by the deadline
- Personal API keys, passwords, or credentials committed to the repository
- Test coverage below 60% (hard minimum)
- Plagiarism: code copied from existing repositories without attribution and modification
- Data leakage: training data contaminating the test set (especially random splits on material data)
- Non-functional Docker build: docker-compose up must start all services without errors
- Missing technical report or README
- Personal forks or copies retained after transfer (verified by Zetheta's security team)

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SECTION F2: SUBMISSION FINAL CHECKLIST

Before executing the repository transfer, verify every item:

- All code committed and pushed to main branch (no dangling branches with unmerged work)
- All tests passing in CI/CD pipeline (green checkmark on latest commit)
- DVC remote configured and all data files pushed
- MLflow experiments exported and committed
- Docker build succeeds without errors (test with docker-compose up --build)
- README.md is complete with architecture diagram and screenshots
- Technical report is complete (15–20 pages, 20+ references)
- Sphinx API documentation generated and committed to docs/
- No personal API keys or credentials in repository (use .env.example template)
- LICENSE file present with Zetheta IP attribution
- .gitignore properly configured (no data files, no __pycache__, no .env, no mlruns/)
- Final commit message: "Final submission – MatRisk AI v1.0 — [Your Email Id] — [Date]"
- Git tag created: git tag -a v1.0 -m "MatRisk AI v1.0 Final Submission"
- Repository to be transferred through Zetheta task pane in the projects ideally – if not possible then Repository transferred via Settings → Danger Zone → Transfer Ownership → ZethetaIntern
- Confirmed repository appears under github.com/ZethetaIntern/matrisk-ai
- All local forks, copies, and clones deleted from personal devices
- Confirmation email sent to project supervisor with transfer timestamp and URL

Post-Transfer: ZethetaIntern will review the repository and score/ rank the submissions.

IP PROTECTION PROTOCOL:

All IP arising from this project including submissions made is the exclusive property of Zetheta Algorithms Pvt. Ltd. as per the confidentiality terms accepted at project commencement

- All innovations documented with dated lab notebook entries in the technical report and git commit history
- Novel algorithms include pseudo-code and mathematical formulation in patent-ready format
- Comparative results against prior art quantified and documented in the model performance report
- The GitHub repository commit history serves as a timestamped invention record under Indian patent law

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END OF PROJECT DOCUMENT